Protocol v3.1: proofs of the cryptographic and differential-privacy claims

Technical appendix: encoding sensitivity, smudging, a worst-case error bound, output privacy (SIM-CDP), input and cohort privacy, correctness

Draft for review

Abstract

This appendix proves, corrects, or marks as open each cryptographic and differential-privacy claim of the specification *Protocol v3.1* (`protocol_v31.tex`). We work in the honest-but-curious model of that specification and reuse its notation. Section 1 proves the encoding-sensitivity lemma. Section 2 proves the smudging lemma for the exact discrete Gaussian, with explicit constants. Section 3 discharges Obligation O1: an explicit bound B_{tot} on the ℓ_2 norm of the whole error ring element, worst case over the clipped data domain, with probability only over keys and fresh randomness. Section 4 extends O1 to decoupled encoding scales and to OpenFHE's uncentred key switching, proves the sensitivity of the centre with decoupled scales, and bounds the decoding offset. Section 5 proves output privacy (Theorem 1) by hybrids. Section 6 proves input and cohort privacy (G1, G2). Section 7 proves the first layer of the disclosure claim G4 (random-oracle model, CDH), bounds the cuckoo-hashing slack, and bounds the reply noise of the PSI in a model of BFV. Section 8 treats correctness, at every level of the circuit. Section 9 proves that Calibrate meets the hypotheses of Theorem 1, that the run computes its DP functionality against R in the standard sense, and that O's work is linear in its scope. Section 10 lists the corrections to the specification.

Summary of results.

- **Lemma 1** holds with the constant $\sqrt{2/N}$ for the stated convention. One precision: Ω must be the *exact* final scale Ω_{fin} of the circuit, not the nominal scale (Correction 2).
- **Lemma 2** holds with the constant $1/\sqrt{2\pi}$ up to an explicit additive smoothing term ($\leq 2^{-98}$ here). An elementary proof gives the constant $1/2$ with no side condition.
- **O1.** The mechanism must be centred at the noiseless circuit output rather than at $\mathsf{Ecd}(\mathbf{z})$ (Correction 1). With that centre, and with every rounding treated adversarially, the circuit exactly as specified gives $B_{\text{tot}} \approx 2^{25.3}$ for stage 2 of the example, with $\delta_B = 2^{-50}$. This needs $\log_2 \Omega \approx 61.5$, not 52.5. The specification's 52.5 floods only the fresh-encryption error, in mean. With the specification's centre, the deterministic plaintext-arithmetic terms alone reach $2^{30.7}$ to $2^{34.5}$, which would need $\log_2 \Omega \approx 67$ to 71. Changes to the circuit bring B_{tot} down to $2^{17.6}$, so that $\log_2 \Omega \approx 53.7$ suffices (Section 3, Table 1):
  - evaluate EvalSum before the first rescale;
  - use stochastic rounding and a single final rescale;
  - make the special modulus larger;
  - give R's encoding a larger scale.
- **Theorem 1** holds under decision-RLWE for two secret distributions, a circular-security assumption for O's rotation-key shares, and O1, with the corrected centring. It holds up to an

explicit statistical slack γ. PSI security is *not* needed for G3 under the specification's adjacency. The (ε, δ) accounting needs the factor $(1+e^{\varepsilon})$ on γ when it is read as indistinguishability-based CDP (Correction 5).

- **G1 and G2** hold under RLWE in s_R (with circular security for R's key material), the random-oracle model for the seed expansion, IND-CPA of the PSI's BFV, and obliviousness of the OPRF. G2 holds up to the cuckoo-hashing failure probability.
- **Correctness.** The q_0 condition in CALIBRATE is sufficient, with a correctness-failure probability of at most $\delta_B + 2Ne^{-50}$ per release, which the specification should state. The exact condition is derived below; no condition is needed at the earlier levels.
- **Added after a second reading of the companion book** (Sections 4, 7 and 9): O1 for OpenFHE's uncentred gadget under model (M3′), with $\varsigma_{\uparrow} = 2\max_j k_j$ and $\varsigma_{\downarrow} = 2k_P$ (Lemma 4.1; the match of OpenFHE 1.5.1 with (M3′) is a code reading); the centre sensitivity with decoupled scales (Lemma 4.3); a public bound on the decoding offset (Proposition 4.5); G4's first layer under CDH in the random-oracle model, without circuit privacy (Theorem 7.3); the soundness of CALIBRATE (Proposition 9.1); standard simulation of R's view jointly with its output (Corollary 9.3); and O's $\Omega(n_{\mathrm{Sc}})$ work (Proposition 9.5). New here: a proved bound on the cuckoo-hashing slack (Lemma 7.5) and a worst-case bound on the PSI's evaluation noise in the textbook BFV model (Proposition 7.7), which covers it by the "+1" of the implementation's B_{err} in that model.

Steps that are not fully rigorous are marked **[Not fully rigorous: ...]**. Citation details we could not verify are marked *[citation detail unverified: ...]*.

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1 Conventions and Lemma 1 (encoding sensitivity)

Ring and embeddings. Let $N \geq 8$ be a power of two, $\mathcal{R} = \mathbb{Z}[X]/(X^N+1)$, $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$, and $\zeta = e^{i\pi/N}$, a primitive $2N$-th root of unity. The roots of X^N+1 are ζ^j for $j \in \mathbb{Z}_{2N}^*$, the odd residues. Let $\mathcal{H} = \langle 5 \rangle \leq \mathbb{Z}_{2N}^*$. Then:

- $\mathcal{H}$ has order $N/2$ and $\mathcal{H} = \{g \in \mathbb{Z}_{2N} : g \equiv 1 \bmod 4\}$;
- $\mathbb{Z}_{2N}^* = \mathcal{H} \sqcup (-\mathcal{H})$;
- the slots are ordered by $h_k = 5^k \bmod 2N$, for $k = 0, \ldots, N/2-1$.

For $f \in \mathbb{R}[X]/(X^N+1)$, the *canonical embedding* is $\sigma(f) = (f(\zeta^j))_{j \in \mathbb{Z}_{2N}^*} \in \mathbb{C}^N$, and the *slot map* is $\tau(f) = (f(\zeta^{h_k}))_{k<N/2} \in \mathbb{C}^{N/2}$. Since f is real, $f(\zeta^{-h}) = \overline{f(\zeta^h)}$. So $\sigma(f)$ consists of $\tau(f)$ and its complex conjugate. The decoding of the specification is $\mathsf{Dcd}(v) = \tau(v)/\Omega$.

We write $\|f\| = \|f\|_2$, $\|f\|_\infty$ and $\|f\|_1$ for the norms of the coefficient vector, and $\|f\|_\infty^{\mathrm{can}} = \|\sigma(f)\|_\infty$. Norms of vectors in $\mathbb{C}^{N/2}$ are complex ℓ_2 norms. For $g \in \mathbb{Z}_{2N}^*$, ψ_g is the automorphism $X \mapsto X^g$.

Lemma 1.1 (Parseval for the canonical embedding)**.** *For every $f \in \mathbb{R}[X]/(X^N+1)$, $\|\sigma(f)\|_2^2 = N\|f\|_2^2$ and $\|\tau(f)\|_2^2 = \frac{N}{2}\|f\|_2^2$. The map τ is an $\mathbb{R}$-linear bijection onto $\mathbb{C}^{N/2}$.*

Proof. We have $\sigma(f) = Wf$ with $W_{j,k} = \zeta^{jk}$, for $j \in \mathbb{Z}_{2N}^*$ and $0 \leq k < N$.

Orthogonality. For $k \neq l$, put $d = l-k$, so that $0 < |d| < N$. Then

$$(W^*W)_{k,l} = \sum_{j \text{ odd}} \zeta^{jd} = \zeta^d \sum_{t=0}^{N-1} \zeta^{2td} = \zeta^d \frac{1-\zeta^{2Nd}}{1-\zeta^{2d}} = 0,$$

because $\zeta^{2Nd} = 1 \neq \zeta^{2d}$. Also $(W^*W)_{k,k} = N$, so $W^*W = N\mathbf{I}$.

Norms. The first identity follows from $W^*W = N\mathbf{I}$. The second follows because $\sigma(f)$ is $\tau(f)$ together with its conjugate.

Bijectivity. Let $y \in \mathbb{C}^N$ satisfy $y_{-j} = \overline{y_j}$. Then $f = N^{-1}W^*y$ has coefficients

$$f_k = N^{-1}\sum_j \zeta^{-jk} y_j, \qquad \overline{f_k} = N^{-1}\sum_j \zeta^{jk}\,\overline{y_j} = N^{-1}\sum_j \zeta^{jk} y_{-j} = f_k,$$

so f is real. Given $z \in \mathbb{C}^{N/2}$, take $y = (z, \bar{z})$ arranged by $\mathcal{H} \sqcup (-\mathcal{H})$. This f satisfies $\tau(f) = z$, and it is unique because W is injective. □

Lemma 1.2 (Encoding sensitivity; Lemma 1 of the specification)**.** *Let $\tau^{-1} : \mathbb{C}^{N/2} \to \mathbb{R}[X]/(X^N+1)$ be the inverse of the slot map.*

(a) $\|\tau^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\,\|\mathbf{z}\|_2$ *for every* $\mathbf{z} \in \mathbb{C}^{N/2}$.

(b) Let $\mathsf{Ecd}_\Omega(\mathbf{z}) = \lfloor \Omega\,\tau^{-1}(\mathbf{z}) \rceil$*, rounded coefficient-wise with any fixed tie rule. For all* $\mathbf{z}, \mathbf{z}'$*,*

$$\|\mathsf{Ecd}_\Omega(\mathbf{z}) - \mathsf{Ecd}_\Omega(\mathbf{z}')\|_2 \le \sqrt{2/N}\,\Omega\,\|\mathbf{z}-\mathbf{z}'\|_2 + \sqrt{N}.$$

(c) Let $\Delta_s = \sup_{D\sim D'} \|\mathbf{z}_s(D) - \mathbf{z}_s(D')\|_2$*, where each released entry occupies exactly one slot and all other slots are zero. Then* $\sup_{D\sim D'} \|\mathsf{Ecd}_\Omega(\mathbf{z}_s(D)) - \mathsf{Ecd}_\Omega(\mathbf{z}_s(D'))\|_2 \le \Delta_{\mathrm{enc},s} = \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$.

Proof. (a) Apply Lemma 1.1 to $f = \tau^{-1}(\mathbf{z})$.

(b) Write $\mathsf{Ecd}_\Omega(\mathbf{z}) = \Omega\tau^{-1}(\mathbf{z}) + \mathbf{r}$ and $\mathsf{Ecd}_\Omega(\mathbf{z}') = \Omega\tau^{-1}(\mathbf{z}') + \mathbf{r}'$, with $\mathbf{r}, \mathbf{r}' \in [-\frac{1}{2}, \frac{1}{2}]^N$. Then $\|\mathbf{r}-\mathbf{r}'\|_\infty \le 1$, so $\|\mathbf{r}-\mathbf{r}'\|_2 \le \sqrt{N}$. Apply (a) to $\mathbf{z}-\mathbf{z}'$, using the $\mathbb{R}$-linearity of τ^{-1}, and the triangle inequality.

(c) Follows from (b). □

Remark 1.3 (Scope of Lemma 1.2)**.**

(i) The constant $\sqrt{2/N}$ belongs to the convention of $N/2$ complex slots measured in complex ℓ_2. Under the full N-dimensional embedding the constant would be $1/\sqrt{N}$. With real data the imaginary parts are zero, so the specification's Δ_s, a real ℓ_2 sensitivity, is the right input.

(ii) The rounding term $\sqrt{N}$ cannot be improved in the worst case. At $N = 2^{14}$ it is 2^7, against $\sqrt{2/N}\,\Omega\Delta_s \approx 2^{52}$.

(iii) The lemma is about $\mathsf{Ecd}_\Omega(\mathbf{z}_s)$. The protocol's decrypted value is centred instead at the output of a noiseless evaluation of the circuit. Section 3 shows that the two differ by a deterministic term of size up to about 2^{30}, and Corollary 3.13 gives the sensitivity of the correct centre. Here Ω must be the *exact* final scale Ω_{fin} (Correction 2).

Facts used in Section 3.

Lemma 1.4 (Norms and automorphisms)**.** *For all* $f, g \in \mathbb{R}[X]/(X^N+1)$*:*

- $\|fg\|_2 \le \|f\|_\infty^{\mathrm{can}}\,\|g\|_2$ *and* $\|fg\|_2 \le \|f\|_1\|g\|_2$*;*
- $\|f\|_2 \le \|f\|_\infty^{\mathrm{can}} \le \|f\|_1 \le \sqrt{N}\|f\|_2$.

Every ψ_g acts as a signed permutation on coefficients and as a permutation on σ. It therefore preserves $\|\cdot\|_2$, $\|\cdot\|_\infty$, $\|\cdot\|_1$ and $\|\cdot\|_\infty^{\mathrm{can}}$.

Proof. Products. We have $\sigma(fg) = \sigma(f) \odot \sigma(g)$. By Lemma 1.1,

$$\|fg\|_2 = N^{-1/2}\|\sigma(f)\odot\sigma(g)\|_2 \le \|f\|_\infty^{\mathrm{can}}\,N^{-1/2}\|\sigma(g)\|_2 = \|f\|_\infty^{\mathrm{can}}\,\|g\|_2.$$

The bound $\|f\|_\infty^{\mathrm{can}} \le \|f\|_1$ holds because $|f(\zeta^j)| \le \sum_k |f_k|$. Also $\|f\|_2 = N^{-1/2}\|\sigma(f)\|_2 \le \|f\|_\infty^{\mathrm{can}}$, and $\|f\|_1 \le \sqrt{N}\|f\|_2$ by Cauchy–Schwarz.

Automorphisms. Take g odd. Then $X^k \mapsto X^{gk \bmod 2N} = \pm X^{gk \bmod N}$, and $k \mapsto gk \bmod N$ is a bijection of $\mathbb{Z}_N$. Moreover $\sigma(\psi_g f)_j = f(\zeta^{jg}) = \sigma(f)_{jg}$. □

Lemma 1.5 (The slot sum is a trace)**.** *Let* $L = \log_2(N/2)$ *and* $g_i = 5^{2^i}$*. For* $0 \le i \le L$*, let* $T_i = \prod_{l=i}^{L-1}(1+\psi_{g_l})$*, so that* $T_i = \sum_{g\in\mathcal{H}_i}\psi_g$ *with* $\mathcal{H}_i = \langle 5^{2^i}\rangle$ *and* $|\mathcal{H}_i| = 2^{L-i}$*. The log-step* EvalSum *evaluates* $\mathrm{A} := T_0 = \sum_{g\in\mathcal{H}}\psi_g$*, and*

$$\mathrm{A}(f) = \tfrac{N}{2}\big(f_0 + f_{N/2}X^{N/2}\big), \qquad \tau(\mathrm{A}f) = \tfrac{N}{2}\big(f_0 + if_{N/2}\big)\mathbf{1} = \Big(\sum_{h\in\mathcal{H}}\sigma(f)_h\Big)\mathbf{1}.$$

Proof. The product formula. Every $g \in \mathcal{H}$ has a unique representation $5^{\sum_i b_i 2^i}$ with $b_i \in \{0,1\}$. This gives the formula for T_i.

The trace of a monomial. By linearity it suffices to treat $f = X^k$. For $j \in \mathcal{H}$ we have $j\mathcal{H} = \mathcal{H}$, so

$$\sigma_j(\mathrm{A}X^k) = S_k := \sum_{g\in\mathcal{H}} \zeta^{gk} = \zeta^k \sum_{t=0}^{N/2-1} \zeta^{4kt}.$$

If $\zeta^{4k} \neq 1$, the geometric sum equals $(1-\zeta^{2Nk})/(1-\zeta^{4k}) = 0$. Now $\zeta^{4k} = 1$ if and only if $k \equiv 0 \pmod{N/2}$. Hence $S_0 = N/2$, $S_{N/2} = \frac{N}{2}\zeta^{N/2} = \frac{N}{2}i$, and $S_k = 0$ otherwise. For $j \in -\mathcal{H}$ the values are the conjugates.

Identification. The polynomial $\frac{N}{2}([k{=}0] + [k{=}N/2]X^{N/2})$ has the same embedding, because $\zeta^{jN/2} = i^j$ equals i on $\mathcal{H}$ and $-i$ on $-\mathcal{H}$. The claim follows from the injectivity of σ. The last expression is $\sigma_j(\mathrm{A}f) = \sum_{g\in\mathcal{H}} \sigma(f)_{jg}$ evaluated at $j = 1$. □

Lemma 1.6 (Masked slot functionals)**.** *Let $f \in \mathbb{R}[X]/(X^N+1)$, $h \in \mathcal{H}$ and $\mathcal{H}' \subseteq \mathcal{H}$. The real-linear map $\xi \mapsto \sigma_h\big(\sum_{g\in\mathcal{H}'} \psi_g(f\xi)\big)$, for $\xi \in \mathbb{R}^N$, equals $\sum_k \xi_k\beta_k$, with*

$$\|\beta\|_2^2 = N \sum_{g\in\mathcal{H}'} |\sigma(f)_{hg}|^2 \le N \min\big\{N\|f\|_2^2,\ |\mathcal{H}'|\,(\|f\|_\infty^{\mathrm{can}})^2\big\}.$$

In particular $\|\beta\|_2^2 = N|\mathcal{H}'|$ when $f = 1$. The real and imaginary parts have coefficient vectors $\Re\beta$ and $\Im\beta$, with $\|\Re\beta\|^2 + \|\Im\beta\|^2 = \|\beta\|^2$.

Proof. We have $\sigma_h(\psi_g(f\xi)) = \sigma(f)_{hg}\sum_k \xi_k\zeta^{hgk}$, so $\beta_k = \sum_{g\in\mathcal{H}'} \sigma(f)_{hg}\zeta^{hgk}$. Expanding,

$$\|\beta\|^2 = \sum_{g,g'\in\mathcal{H}'} \sigma(f)_{hg}\overline{\sigma(f)_{hg'}} \sum_{k<N} \zeta^{hk(g-g')}.$$

If $g \neq g'$, then $m = h(g-g')$ is even and $m \not\equiv 0 \pmod{2N}$, so ζ^m is an N-th root of unity other than 1 and the inner sum vanishes. If $g = g'$, the inner sum is N. The bounds follow from Lemma 1.1 and from $|\sigma(f)_{hg}| \le \|f\|_\infty^{\mathrm{can}}$. □

Lemma 1.7 (Masking)**.** *Let $\mathbf{u}_k$ be the one-hot slot vector and $m_{u_k} = \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_k) = \Omega_u\tau^{-1}(\mathbf{u}_k) + r_{u_k}$, with $\|r_{u_k}\|_\infty \le \frac{1}{2}$. For polynomials y_e and distinct slots k_e:*

$$\Big\|\sum_e y_e\,\Omega_u\tau^{-1}(\mathbf{u}_{k_e})\Big\|_2 = \Omega_u\sqrt{2/N}\Big(\sum_e |\tau(y_e)_{k_e}|^2\Big)^{1/2}, \qquad \|y_e\,r_{u_k}\|_2 \le \tfrac{N}{2}\|y_e\|_2.$$

Proof. The slot map is multiplicative, so $\tau(y_e\tau^{-1}(\mathbf{u}_{k_e})) = \tau(y_e)_{k_e}\mathbf{u}_{k_e}$. Apply Lemma 1.2(a) to the sum, whose slots are disjoint. The second bound is Lemma 1.4 with $\|r_{u_k}\|_1 \le N/2$. □

2 Lemma 2 (smudging with the exact discrete Gaussian)

Conventions. For $\sigma > 0$ and $x \in \mathbb{R}^N$, let $\rho_\sigma(x) = \exp(-\|x\|^2/(2\sigma^2))$. For a countable set S, write $\rho_\sigma(S) = \sum_{x\in S}\rho_\sigma(x)$. For $c \in \mathbb{R}^N$, $D_{\mathbb{Z}^N+c,\sigma}$ is the distribution on $\mathbb{Z}^N + c$ with probability mass function $\rho_\sigma(x)/\rho_\sigma(\mathbb{Z}^N + c)$. The specification's $\mathcal{N}_\mathbb{Z}(0,\sigma^2)^N$ is $D_{\mathbb{Z}^N,\sigma}$, the product of N independent copies of the one-dimensional discrete Gaussian $\mathcal{N}_\mathbb{Z}(0,\sigma^2)$ of Canonne, Kamath and Steinke [4]. SD is statistical (total-variation) distance.

Smoothing parameters are usually stated with the Gaussian width $s = \sqrt{2\pi}\sigma$, where $\rho_\sigma(x) = e^{-\pi\|x\|^2/s^2}$. For a lattice Λ and $\epsilon > 0$, $\eta_\epsilon(\Lambda)$ is the least s such that $\sum_{y\in\Lambda^*\setminus\{0\}} e^{-\pi s^2\|y\|^2} \le \epsilon$ [13].

Lemma 2.1 (Gaussian sums are maximal at the lattice)**.** *For all $\sigma > 0$ and $c \in \mathbb{R}^N$, $\rho_\sigma(\mathbb{Z}^N + c) \le \rho_\sigma(\mathbb{Z}^N)$. Equality holds if $c \in \mathbb{Z}^N$.*

Proof. The sum factorizes over coordinates, so it suffices to take $N = 1$. By Poisson summation for the Gaussian,

$$\sum_{z\in\mathbb{Z}} e^{-(z+c)^2/(2\sigma^2)} = \sqrt{2\pi}\,\sigma \sum_{k\in\mathbb{Z}} e^{-2\pi^2\sigma^2k^2}\cos(2\pi kc) \le \sqrt{2\pi}\,\sigma \sum_{k\in\mathbb{Z}} e^{-2\pi^2\sigma^2k^2} = \rho_\sigma(\mathbb{Z}).$$

For integer c the sum is a re-indexing of $\rho_\sigma(\mathbb{Z})$. □

Lemma 2.2 (Divergences of an integer shift)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$, $\mathbf{e} \in \mathbb{Z}^N$, and let $P_\mathbf{e}$ be the law of $\eta + \mathbf{e}$. For every $\alpha > 1$,*

$$D_\alpha(P_\mathbf{e}\|P_\mathbf{0}) \le \frac{\alpha\|\mathbf{e}\|^2}{2\sigma^2} \qquad \text{and} \qquad \mathrm{KL}(P_\mathbf{e}\|P_\mathbf{0}) = \frac{\|\mathbf{e}\|^2}{2\sigma^2}.$$

Proof. Normalizers. Since $\mathbf{e}$ is integral, $P_\mathbf{e}(y) = \rho_\sigma(y-\mathbf{e})/Z$ and $P_\mathbf{0}(y) = \rho_\sigma(y)/Z$ with the same $Z = \rho_\sigma(\mathbb{Z}^N)$.

Rényi divergence. Use the identity $\alpha\|y-\mathbf{e}\|^2 + (1-\alpha)\|y\|^2 = \|y-\alpha\mathbf{e}\|^2 + \alpha(1-\alpha)\|\mathbf{e}\|^2$. With Lemma 2.1,

$$\sum_y P_\mathbf{e}(y)^\alpha P_\mathbf{0}(y)^{1-\alpha} = \frac{\rho_\sigma(\mathbb{Z}^N - \alpha\mathbf{e})}{Z}\, e^{\alpha(\alpha-1)\|\mathbf{e}\|^2/(2\sigma^2)} \le e^{\alpha(\alpha-1)\|\mathbf{e}\|^2/(2\sigma^2)}.$$

Take $\frac{1}{\alpha-1}\ln$ of both sides.

Kullback–Leibler divergence. By symmetry $\mathbb{E}\eta = \mathbf{0}$, so

$$\mathrm{KL}(P_\mathbf{e}\|P_\mathbf{0}) = \mathbb{E}\big[\|\eta+\mathbf{e}\|^2 - \|\eta\|^2\big]/(2\sigma^2) = \big(2\mathbb{E}\langle\eta,\mathbf{e}\rangle + \|\mathbf{e}\|^2\big)/(2\sigma^2) = \|\mathbf{e}\|^2/(2\sigma^2).$$

□

Lemma 2.3 (Smudging; Lemma 2 of the specification, with explicit constants)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ and $\mathbf{e} \in \mathbb{Z}^N$.*

(a) Elementary bound, valid for all $\sigma > 0$: $\mathrm{SD}(\eta+\mathbf{e},\eta) \le \|\mathbf{e}\|_2/(2\sigma)$.

(b) Sharp constant: *let $\epsilon \in (0,1)$, $\sigma_2 \ge \eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi}$, $\sigma > \sigma_2$ and $\sigma_1 = \sqrt{\sigma^2 - \sigma_2^2}$. Then*

$$\mathrm{SD}(\eta+\mathbf{e},\eta) \le \frac{\|\mathbf{e}\|_2}{\sqrt{2\pi}\,\sigma_1} + \frac{4\epsilon}{1-\epsilon}.$$

By [13, Lemma 3.3], $\eta_\epsilon(\mathbb{Z}^N) \le \sqrt{\ln(2N(1+1/\epsilon))/\pi}$. For $N = 2^{14}$ and $\epsilon = 2^{-100}$ this gives $\eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi} \le 2.01$, so $\sigma_2 = 2.1$ is admissible.

Proof. (a) Apply Pinsker's inequality, $\mathrm{SD} \le \sqrt{\mathrm{KL}/2}$, to Lemma 2.2.

(b) The proof constructs a shift-equivariant post-processing F of a continuous Gaussian whose output is close to η.

The post-processing. Let $x \sim \mathcal{N}(0,\sigma_1^2\mathbf{I}_N)$ be continuous. Let $F(x) = x + z'$, where $z' \sim D_{\mathbb{Z}^N - x,\sigma_2}$ is sampled given x. Then $F(x) \in \mathbb{Z}^N$. For $\mathbf{e} \in \mathbb{Z}^N$ we have $\mathbb{Z}^N - x - \mathbf{e} = \mathbb{Z}^N - x$, so $F(x+\mathbf{e})$ and $F(x) + \mathbf{e}$ have the same law.

$F(x)$ is close to η. The law of $F(x)$ is

$$P(y) = \int \phi_{\sigma_1}(x)\,\frac{\rho_{\sigma_2}(y-x)}{\rho_{\sigma_2}(\mathbb{Z}^N - x)}\,dx, \qquad y \in \mathbb{Z}^N.$$

Since $\sqrt{2\pi}\sigma_2 \geq \eta_\epsilon(\mathbb{Z}^N)$, the smoothing lemma [13, Lemma 4.4] gives $\rho_{\sigma_2}(\mathbb{Z}^N - x) \in [\frac{1-\epsilon}{1+\epsilon}, 1]\,\rho_{\sigma_2}(\mathbb{Z}^N)$ for every x. The Gaussian convolution identity gives $\int \phi_{\sigma_1}(x)\rho_{\sigma_2}(y-x)\,dx = (\sigma_2/\sigma)^N \rho_\sigma(y)$. Hence $P(y) = C\,\kappa(y)\rho_\sigma(y)$, with $\kappa(y) \in [1, r]$ and $r = \frac{1+\epsilon}{1-\epsilon}$. Normalizing, $P(y)/D_{\mathbb{Z}^N,\sigma}(y) \in [1/r, r]$, and therefore $\mathrm{SD}(F(x), \eta) \leq r - 1 = 2\epsilon/(1-\epsilon)$.

The shift. By the triangle inequality and the data-processing inequality,

$$\mathrm{SD}(\eta + \mathbf{e}, \eta) \leq \mathrm{SD}(F(x+\mathbf{e}), F(x)) + \frac{4\epsilon}{1-\epsilon} \leq \mathrm{SD}(x+\mathbf{e}, x) + \frac{4\epsilon}{1-\epsilon}.$$

By rotation invariance, $\mathrm{SD}(x+\mathbf{e}, x) = 2\Phi(\|\mathbf{e}\|/(2\sigma_1)) - 1 \leq \|\mathbf{e}\|/(\sqrt{2\pi}\sigma_1)$, since $\Phi' \leq 1/\sqrt{2\pi}$. □

Corollary 2.4 (Flooding with side information)**.** *Let $(Z, \mathbf{e})$ have any joint law, with $\mathbf{e} \in \mathbb{Z}^N$. Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ be independent of $(Z, \mathbf{e})$, and let $B > 0$. Then*

$$\mathrm{SD}\big((Z, \mathbf{e}+\eta), (Z, \eta)\big) \leq \Pr[\|\mathbf{e}\|_2 > B] + \sup_{\|\mathbf{e}'\|_2 \leq B} \mathrm{SD}(\mathbf{e}'+\eta, \eta).$$

The same bound holds after reducing $\mathbf{e}+\eta$ and η modulo q_0, by data processing. In particular, suppose $\sigma \geq \max\{2^\kappa B/\sqrt{2\pi},\, 2^{42}\}$, with $\epsilon = 2^{-100}$ and $\sigma_2 = 2.1$ as above. Then the right-hand side is at most

$$\Pr[\|\mathbf{e}\| > B] + 2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9} \leq \Pr[\|\mathbf{e}\| > B] + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}.$$

Proof. Condition on $(Z, \mathbf{e}) = (z, \mathbf{e}')$. Because η is independent of $(Z, \mathbf{e})$, the conditional distance is $\mathrm{SD}(\mathbf{e}'+\eta, \eta)$. This is at most the supremum when $\|\mathbf{e}'\| \leq B$, and at most 1 otherwise. Averaging over $(z, \mathbf{e}')$ gives the bound. For the numerical form, Lemma 2.3(b) gives $\mathrm{SD}(\mathbf{e}'+\eta, \eta) \leq \|\mathbf{e}'\|/(\sqrt{2\pi}\sigma_1) + 4\epsilon/(1-\epsilon) \leq 2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9}$. Moreover $\sigma/\sigma_1 = (1-\sigma_2^2/\sigma^2)^{-1/2} \leq 1 + \sigma_2^2/\sigma^2$ whenever $\sigma \geq 2\sigma_2$, and for $\sigma \geq 2^{42}$ this is at most $1 + 4.41 \cdot 2^{-84} < 1 + 2^{-81}$. Hence $2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9} \leq 2^{-\kappa}(1+2^{-80}) + 2^{-97}$. □

Remark 2.5 (The floor 2^{42}; corrected)**.** An earlier version of this corollary stated the final inequality, in the form $\leq \Pr[\|\mathbf{e}\| > B] + 2^{-\kappa} + 2^{-97}$, "for every $\sigma \geq 2^{20}$". That is false for $\kappa < 60$: at $\sigma = 2^{20}$ the factor σ/σ_1 is $1 + 2^{-38.9}$, and the proof's bound $1 + \sigma_2^2/\sigma^2$ is $1 + 2^{-37.9}$, so the excess $2^{-\kappa} \cdot 2^{-37.9}$ exceeds the room $2^{-97} - 2^{-97.9}$ unless $\kappa \geq 60$. The factor $1 + 2^{-80}$ needs $\sigma \geq 2.1 \cdot 2^{40} \approx 2^{41.1}$; the floor is now 2^{42} (composition appendix, Remark 2.6). Nothing numerical changes: at every evaluation setting $\sigma_c \geq 2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi} \geq 2^{55}$. The hockey-stick form of this step (composition appendix, Lemma 5.1) needs no floor at all.

Remark 2.6 (Comparison with the specification)**.** The specification states Lemma 2 with "$\sigma_c \gg 1$" and "+negl". Lemma 2.3(b) makes both precise:

- the condition is $\sigma_c > \sigma_2 \approx 2.1$;
- the negligible term is $4\epsilon/(1-\epsilon) \leq 2^{-97}$;
- the per-release distance is $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ rather than exactly $2^{-\kappa}$.

The difference is immaterial, but the δ accounting should use the corrected value (Correction 5). An implementation may prefer the elementary bound (a), which needs no smoothing argument. The condition then becomes $\sigma_c \geq 2^\kappa B_{\mathrm{tot}}/2$, which costs $\log_2(\sqrt{2\pi}/2) \approx 0.33$ bits of Ω.

3 Obligation O1: a worst-case bound on the whole error

The review requires the bound to be worst case over the data, for two reasons. CKKS decryptions leak the secret through their error [12]. And flooding calibrated to average-case noise estimates has been broken by adversarially chosen inputs [9]. Below, every polynomial determined by the data, or by the public randomness, is treated adversarially. Probabilities are taken only over secrets, errors and fresh coins, which are independent of those polynomials (Lemma 3.6).

3.1 Probability tools

A real random variable X is *v-subgaussian* if $\mathbb{E}e^{\lambda X} \le e^{v\lambda^2/2}$ for all $\lambda \in \mathbb{R}$.

Lemma 3.1 (Variance proxies)**.** *(a) The uniform distribution on $\{-1, 0, 1\}$ is $\frac{2}{3}$-subgaussian. Sums of independent variables add proxies, so each coefficient of $s = s_R + s_O$ is $\frac{4}{3}$-subgaussian.*
(b) $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ is σ^2-subgaussian.
(c) A mean-zero variable supported on an interval of length 1 is $\frac{1}{4}$-subgaussian.

Proof. (a) Expand in powers of λ: $\frac{1}{3} + \frac{2}{3}\cosh\lambda = 1 + \sum_{k\ge1} \frac{2}{3(2k)!}\lambda^{2k}$ and $e^{\lambda^2/3} = 1 + \sum_{k\ge1} \frac{\lambda^{2k}}{3^k k!}$. Compare the series term by term, using $\frac{2}{3}\,3^k k! \le (2k)!$ for all $k \ge 1$.

(b) Complete the square:

$$\mathbb{E}e^{\lambda X} = \frac{\sum_z e^{-z^2/(2\sigma^2)+\lambda z}}{\rho_\sigma(\mathbb{Z})} = e^{\lambda^2\sigma^2/2}\,\frac{\rho_\sigma(\mathbb{Z} - \lambda\sigma^2)}{\rho_\sigma(\mathbb{Z})} \le e^{\lambda^2\sigma^2/2},$$

by Lemma 2.1.

(c) This is Hoeffding's lemma. □

Lemma 3.2 (Hsu, Kakade and Zhang [11], Theorem 1)**.** *Let $x \in \mathbb{R}^n$ satisfy $\mathbb{E}e^{\alpha^\top x} \le e^{\|\alpha\|^2/2}$ for all α. This holds, for example, when the coordinates are independent and 1-subgaussian. Let $A \in \mathbb{R}^{m\times n}$ and $\Sigma = A^\top A$. Then, for all $t > 0$,*

$$\Pr\big[\|Ax\|^2 > \operatorname{tr}\Sigma + 2\sqrt{\operatorname{tr}(\Sigma^2)t} + 2\|\Sigma\|t\big] \le e^{-t}.$$

Since $\operatorname{tr}(\Sigma^2) \le \|\Sigma\|\operatorname{tr}\Sigma$, it follows that $\|Ax\|_2 \le \|A\|_F + \sqrt{2t}\,\|A\|_{\mathrm{op}}$ with probability at least $1 - e^{-t}$.

Corollary 3.3 (Products with fixed polynomials)**.** *Let $\xi_1, \dots, \xi_m \in \mathbb{Z}^N$ be independent random vectors with independent coordinates, where those of ξ_j are mean-zero and v_j-subgaussian. Let $f_1, \dots, f_m$ be fixed polynomials, or random but independent of the ξ_j. Then, with probability at least $1 - e^{-t}$,*

$$\Big\|\sum_j f_j\xi_j\Big\|_2 \le \Big(N\sum_j v_j\|f_j\|_2^2\Big)^{1/2} + \Big(2t\sum_j v_j(\|f_j\|_\infty^{\mathrm{can}})^2\Big)^{1/2}.$$

More generally, for any fixed real-linear map $\ell : (\mathbb{R}^N)^m \to \mathbb{R}^d$ with matrix A in the scaled variables $\xi_j/\sqrt{v_j}$, $\|\ell(\xi)\| \le \|A\|_F(1 + \sqrt{2t})$ with probability at least $1 - e^{-t}$.

Proof. Multiplication by f has the matrix $M_f = N^{-1}W^*\mathrm{diag}(\sigma(f))W$. This matrix is normal, so $\|M_f\|_F^2 = \|\sigma(f)\|^2 = N\|f\|^2$ and $\|M_f\|_{\mathrm{op}} = \|f\|_\infty^{\mathrm{can}}$. Apply Lemma 3.2 to the block matrix $[\sqrt{v_1}M_{f_1} \mid \cdots \mid \sqrt{v_m}M_{f_m}]$. Its operator norm satisfies $\|A\|_{\mathrm{op}}^2 \le \sum_j v_j\|M_{f_j}\|_{\mathrm{op}}^2$. For the general form, use $\|A\|_{\mathrm{op}} \le \|A\|_F$. When the f_j are random and independent of the ξ_j, condition on them. □

Lemma 3.4 (Norms of random polynomials)**.** *Let $\xi \in \mathbb{Z}^N$ have independent, mean-zero, v-subgaussian coordinates. Then:*

- *$\|\xi\|_2 \le \sqrt{v}(\sqrt{N} + \sqrt{2t})$ with probability at least $1 - e^{-t}$;*
- *$\|\xi\|_\infty^{\mathrm{can}} \le \sqrt{2vN\ln(2N/\delta)}$ with probability at least $1 - \delta$.*

Proof. The first bound is Lemma 3.2 with $A = \mathbf{I}$.

For the second, take h odd. The real part $\Re\sigma(\xi)_h = \sum_k \xi_k\cos(\pi hk/N)$ is $vN/2$-subgaussian, because $\sum_{k<N}\cos^2(\pi hk/N) = N/2$; the same holds for the imaginary part. So each part exceeds a in absolute value with probability at most $2e^{-a^2/(vN)}$. There are $N/2$ conjugate pairs of slots and two parts each, so a union bound gives $\|\xi\|_\infty^{\mathrm{can}} \le \sqrt{2}\,a$ except with probability $2Ne^{-a^2/(vN)}$. □

3.2 The circuit, made explicit

The specification fixes the circuit only up to standard choices for CKKS [7], its RNS variant [8] and hybrid key switching [10]. We make the following choices explicit; Section 10 discusses deviations from them.

(M1) *Moduli.* Ciphertexts start at $Q_2 = q_0q_1q_2$, where q_0 may be a product of RNS primes. P is the special modulus. $\textsc{Rescale}_q$ maps $c \mapsto (c - [c]_q)/q$ component-wise, as RNS-CKKS does. Its rounding error on each component lies in $[-\frac{1}{2}, \frac{1}{2}]^N$.

(M2) *Scales.* R encodes its vectors at scale Ω_U. O encodes V_e at Ω_V and the masks at Ω_u. The final scale $\Omega_{\text{fin}} := \Omega_U\Omega_V\Omega_u/(q_1q_2)$ is public, and R decodes with Ω_{fin}. The example uses $\Omega_U = q_2$ and $\Omega_V = \Omega_u = q_1 = \Omega_{\text{fin}} = \Omega$.

(M3) *Key switching*, in the hybrid form.

- *Gadget.* The primes of $Q_1 = q_0q_1$ are partitioned into d_{num} digits with products $\tilde{Q}_j$. Let g_j be the CRT idempotent of digit j: it is $\equiv 1$ modulo the primes of digit j and $\equiv 0$ modulo the others.
- *Decomposition.* $\tilde{D}_j(c) \in \mathcal{R}$ satisfies $\tilde{D}_j(c) \equiv c \pmod{\tilde{Q}_j}$ and $\|\tilde{D}_j(c)\|_\infty \le \varsigma_\uparrow \tilde{Q}_j/2$. Here $\varsigma_\uparrow = 1$ for exact ModUp, and $\varsigma_\uparrow \le k_j$, the number of primes in digit j, for fast base extension.
- *Joint rotation keys.* The key for ψ_g is $\text{rk}_j = (b_j, a_j)$, with $b_j = -a_js + P\mathsf{g}_j\psi_g(s) + e_j^{\text{rk}} \bmod PQ_2$. Here a_j is a common random string and $e_j^{\text{rk}} = e_{R,j}^{\text{rk}} + e_{O,j}^{\text{rk}}$. This is the sum of the two parties' shares $-a_js_P + P\mathsf{g}_j\psi_g(s_P) + e_{P,j}^{\text{rk}}$ in the protocol of Mouchet et al. [17]. It is used at level 1 after reduction modulo PQ_1.
- *The switch.* $\text{KS}(c_0, c_1) = \big(c_0 + \lfloor P^{-1}\sum_j \tilde{D}_j(c_1)b_j\rceil,\ \lfloor P^{-1}\sum_j \tilde{D}_j(c_1)a_j\rceil\big)$. The ModDown rounding errors rd_0, rd_1 lie in $[-\frac{1}{2}, \frac{1}{2}]^N$.

(M4) *Release* is Algorithm 5 at level 0, with $p_0 = -as + e^{\text{pk}} \bmod q_0$ and $e^{\text{pk}} = e_R^{\text{pk}} + e_O^{\text{pk}}$. After two rescales the ciphertext is at q_0. The "modulus switch to q_0" of Algorithm 4 is therefore a no-op, or an exact limb drop.

(M5) *Distributions.* $s_R, s_O, u \leftarrow \chi_{\text{key}}$, uniform ternary. Every error is drawn from $\chi_{\text{err}} = \mathcal{N}_{\mathbb{Z}}(0, \sigma_e^2)^N$, with $\sigma_e = 3.2$.

Lemma 3.5 (Key switching)**.** *Suppose* $c_0 + c_1\psi_g(s) \equiv \mu \pmod{Q_1}$. *Then* $\text{KS}(c_0, c_1) = (c_0', c_1')$ *satisfies* $c_0' + c_1's \equiv \mu + e_{\text{ks}} \pmod{Q_1}$, *where*

$$e_{\text{ks}} = P^{-1}\sum_j \tilde{D}_j(c_1)\, e_j^{\text{rk}} + \text{rd}_0 + \text{rd}_1 s.$$

Proof. Modulo PQ_1, $\sum_j \tilde{D}_j(b_j + a_js) = P\sum_j \tilde{D}_j\mathsf{g}_j\psi_g(s) + \sum_j \tilde{D}_j e_j^{\text{rk}}$. By the CRT, $\sum_j \tilde{D}_j(c_1)\mathsf{g}_j \equiv c_1 \pmod{Q_1}$. Divide by P, which is exact modulo Q_1 for multiples of P, and add the rounding errors. □

Lemma 3.6 (The c_1-components are independent of all secrets and errors)**.** *At every step of Algorithm 4, the* c_1*-component of every ciphertext is a deterministic function of three things: the seeds' expansions* a_h*, the common random strings* a_j *of the rotation keys, and the data. Consequently* $\tilde{D}_j(\cdot)$, rd_1 *and the* c_1*-rounding errors of every rescale are independent of* $(s_R, s_O, u,$ *all errors*$)$.

Proof. By induction over the operations:

- the conversion leaves $c_1 = a_h$;
- a ct–pt product maps $c_1 \mapsto m_V c_1$, and additions add c_1-components;
- a rescale maps $c_1 \mapsto \lfloor c_1/q\rceil$;
- an automorphism maps $c_1 \mapsto \psi_g(c_1)$;
- key switching outputs $\lfloor P^{-1}\sum_j \tilde{D}_j(c_1)a_j\rceil$.

None of these uses c_0, a secret or an error. □

3.3 The noiseless centre and the error decomposition

For entry e, let U_e be R's vector and V_e O's vector. Define the plaintexts $m_{U_e,h} = \mathsf{Ecd}_{\Omega_U}(U_e[h])$, $m_{V_e,h} = \mathsf{Ecd}_{\Omega_V}(V_e[h])$ and $m_{u_e} = \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_{k_e}) = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$.

Definition 3.7 (Noiseless evaluation)**.** For release s,

$$\widehat{Z}_s(D) = \sum_{e \in \mathcal{E}_s} \frac{1}{q_1 q_2} \mathrm{A}\Big(\sum_{h \le m} m_{U_e,h} m_{V_e,h}\Big) m_{u_e} \ \in \mathbb{Q}[X]/(X^N+1), \qquad f_s(D) = \lfloor \widehat{Z}_s(D) \rceil \in \mathbb{Z}^N.$$

This is a deterministic function of O's data, R's data and the public parameters. It does not depend on any key, error or random coin.

Proposition 3.8 (Error decomposition)**.** *Let $v_s = h_{O,s} + c'_{1,s} s_R \bmod q_0$. Under (M1)–(M5),*

$$v_s \equiv f_s(D) + e_{\mathrm{tot},s} + \eta_s \pmod{q_0}, \qquad e_{\mathrm{tot},s} = \sum_{i=1}^{8} T_i \in \mathbb{Z}^N,$$

where, writing $\Pi_e(y) := q_1^{-1} y\, \Omega_u \tau^{-1}(\mathbf{u}_{k_e})$ for the main part of masking and rescaling,

$$\begin{aligned}
T_1 &= \sum_e \Pi_e\big(\mathrm{A}(x_e^R)\big), \quad x_e^R = q_2^{-1} \sum_h e_{R,h}^{(U_e)} m_{V_e,h} && \text{(R's encryption errors)}\\
T_2 &= \sum_e \Pi_e\big(\mathrm{A}(\varepsilon_{0,e} + \varepsilon_{1,e} s)\big) && \text{(first rescale)}\\
T_3 &= \sum_e \Pi_e\Big(\sum_{i<L} T_{i+1}\big(P^{-1} \textstyle\sum_j \tilde{D}_{e,i,j} e_j^{\mathrm{rk},(i)}\big)\Big) && \text{(key-switching gadget)}\\
T_4 &= \sum_e \Pi_e\Big(\sum_{i<L} T_{i+1}\big(\mathrm{rd}_{0,e,i} + \mathrm{rd}_{1,e,i} s\big)\Big) && \text{(ModDown roundings)}\\
T_5 &= \sum_e (\varepsilon'_{0,e} + \varepsilon'_{1,e} s) && \text{(second rescale)}\\
T_6 &= e^{\mathrm{pk}} u + e_0 + e_1 s && \text{(fresh encryption)}\\
T_7 &= \widehat{Z}_s(D) - f_s(D), \quad \|T_7\|_\infty \le \tfrac{1}{2} && \text{(rounding of the centre)}\\
T_8 &= \sum_e q_1^{-1} Y_e\, r_{u_e} && \text{(mask spill of the errors).}
\end{aligned}$$

Here $Y_e = \mathrm{A}(x_e^R + \varepsilon_{0,e} + \varepsilon_{1,e} s) + \sum_i T_{i+1}(e_{\mathrm{ks},e,i})$. All ε and rd *vectors lie in $[-\frac{1}{2}, \frac{1}{2}]^N$, and $e_j^{\mathrm{rk},(i)}$ is the error of digit j of the key for ψ_{g_i}.*

Proof. Track the decryption invariant $c_0 + c_1 s$, as integer polynomials, through Algorithm 4.

Conversion and products. The conversion gives $c_0 + a_h s = m_{U,h} + e_{R,h}$ exactly. After the products and the chunk sum, the value is $\mu_{\mathbb{Z}} + KQ_2$ for some $K \in \mathcal{R}$, with $\mu_{\mathbb{Z}} = \sum_h m_{U,h} m_{V,h} + \sum_h e_{R,h} m_{V,h}$.

First rescale. Write the rounding of c_j/q_2 as $c_j/q_2 + \varepsilon_j$. Then $c'_0 + c'_1 s = \mu_{\mathbb{Z}}/q_2 + \varepsilon_0 + \varepsilon_1 s + KQ_1$.

EvalSum. By Lemma 3.5 and $\psi_g(c_0) + \psi_g(c_1)\psi_g(s) = \psi_g(c_0 + c_1 s)$, induction over the L steps gives $\mathrm{A}(\mu^{(2)}) + \sum_i T_{i+1}(e_{\mathrm{ks},i}) \pmod{Q_1}$, by Lemma 1.5.

Masking and second rescale. Multiplying by m_{u_e} and rescaling by q_1 adds $\varepsilon'_0 + \varepsilon'_1 s$ modulo q_0.

Release. Summing over e, the release adds $(-as + e^{\mathrm{pk}})u + e_0 + (au + e_1)s + \eta = e^{\mathrm{pk}} u + e_0 + e_1 s + \eta$.

Collecting terms. Split $m_{u_e} = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$. The terms not involving any key, error or rounding make up exactly $\widehat{Z}_s(D)$; the rest are $T_1, \dots, T_6$ and T_8. Since v_s, f_s and η_s are integral, so is $e_{\mathrm{tot},s}$. □

Remark 3.9 (Why the centre matters)**.** The specification centres the mechanism at $\mathsf{Ecd}_\Omega(\mathbf{z}_s(D))$ and counts the encoding roundings of V_e and $\mathbf{u}_k$ as error. The deterministic difference $\widehat{Z}_s(D) - \Omega\tau^{-1}(\mathbf{z}_s(D))$ contains the following terms:

- the *mask spill* of the statistic itself, $\sum_e(\Omega_U\Omega_V/(q_1q_2))S_e\, r_{u_e}$, where S_e is the entry's value. Its ℓ_2 norm is about $\sqrt{N/12}\,(\sum_e S_e^2)^{1/2}$. This is up to $2^{30.7}$ for $\bar{n}_O = 10^6$, and up to $2^{34.5}$ by the triangle inequality;
- the cross terms $\mathrm{A}(r_U \cdot \Omega_V\tau^{-1}(V))$ between R's rounding and O's data. These are up to $\frac{\sqrt{N}}{2}\sqrt{m}\,(\bar{n}_O\Gamma_{\mathcal{E}})^{1/2} \approx 2^{25}$, in the notation of Section 3.4;
- a nominal-versus-exact scale mismatch of relative size about 2^{-30} when the implementation decodes with a nominal scale, as OpenFHE's FIXEDMANUAL does. This gives about 2^{23}; see Section 3.6.

All three are deterministic functions of D. They are part of the *query*, not hints about secrets. They need no flooding, provided the mechanism is centred at $f_s(D)$ and the sensitivity of f_s is used (Corollary 3.13). Flooding them instead would require $B_{\mathrm{tot}} \approx 2^{30.7}$ to $2^{34.5}$, and hence $\log_2\Omega \approx 67$ to 71 for stage 2 of the example (script `scratch/ecd_centre.py`).

3.4 Term-by-term bounds, worst case over data

Let $|\mathcal{E}|$ be the number of entries in the release and $m = \lceil \bar{n}_O/N_s \rceil$. Fix δ_B and split it into $n_{\mathrm{ev}} = 12$ equal parts. Put

$$t = \ln(n_{\mathrm{ev}}/\delta_B), \qquad \tau_t = 1 + \sqrt{2t}, \qquad \ell_N = \ln(2Nn_{\mathrm{ev}}/\delta_B), \qquad v_s = \tfrac{4}{3}, \quad v_u = \tfrac{2}{3}.$$

Lemma 3.10 (Data bounds)**.** *Every dataset in the clipped domain satisfies* $\sum_{e\in\mathcal{E}} V_{e,t}^2 \le \Gamma_{\mathcal{E}}$ *for every row* t*, where*

$$\Gamma_{\mathcal{E}_G} = o^2 + (d_R+1)o, \qquad \Gamma_{\mathcal{E}_2} = o^2 + (d_R+1)(o+w) + ow + w^2.$$

Consequently $\sum_e\sum_h \|m_{V_e,h}\|_2^2 \le M_V^2$*, with* $M_V := \Omega_V\sqrt{2/N}\,(\bar{n}_O\Gamma_{\mathcal{E}})^{1/2} + \frac{1}{2}(|\mathcal{E}|mN)^{1/2}$.

Proof. Bound the entries one block at a time:

- for $\mathrm{triu}\,\mathbf{B}$: $\sum_{j\le k}(x_jx_k)^2 = \frac{1}{2}(\|\mathbf{x}\|^4 + \sum_j x_j^4) \le \|\mathbf{x}\|^4 \le o^2$;
- for $\mathbf{C}_{jk}$, where $V = x_{O,k}$ for d_R+1 values of j: $(d_R+1)\|\mathbf{x}\|^2$;
- for $\mathbf{c}_R$: $(d_R+1)y^2$;
- for $\mathbf{c}_O$: $\|\mathbf{x}\|^2y^2$;
- for $\mathbf{y}^\top\mathbf{y}$: y^4.

Dummy and unmatched rows have $V = 0$, or are bounded in the same way. For the second claim, $\|m_{V,h}\| \le \Omega_V\sqrt{2/N}\|V_h\| + \sqrt{N}/2$ by Lemma 1.2. Then apply Minkowski's inequality over the family (e,h), and use $\sum_{e,h}\|V_{e,h}\|^2 \le \bar{n}_O\Gamma_{\mathcal{E}}$. □

Proposition 3.11 (Bounds on the terms)**.** *Assume (M1)–(M5), and let* $\tilde{Q}_{\max} = \max_j \tilde{Q}_j$*. Fix all public randomness: the expansions* a_h*, the common random strings* a, a_j*, and the dataset* D *in the clipped domain. Then each of the following bounds fails with probability at most* δ_B/n_{ev} *over*

$(s_R, s_O, u, \textit{all errors})$*; the bound on* T_6 *uses five of these events.*

$$\begin{aligned}
\|T_1\| &\le B_1 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\tfrac{N}{2q_2}\,\sqrt{2}\,\sigma_e\,M_V\,\tau_t,\\
\|T_2\| &\le B_2 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\tfrac{N}{2}\Big(\sqrt{|\mathcal{E}|/2} + \sqrt{v_s}\sqrt{2|\mathcal{E}|}\,\tfrac{\sqrt{N}}{2}\,\tau_t\Big),\\
\|T_3\| &\le B_3 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\sqrt{2}\,\sigma_e\,\big(|\mathcal{E}|\,L\,d_{\mathrm{num}}\big)^{1/2}\,\frac{\varsigma_\uparrow N^{3/2}\tilde{Q}_{\max}}{2P}\,\tau_t,\\
\|T_4\| &\le B_4 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\sqrt{|\mathcal{E}|}\Big(\tfrac{N}{2}\sum_{i<L} 2^{(L-1-i)/2} + \sqrt{v_s}\,L\,\tfrac{N^{3/2}}{2}\,\tau_t\Big),\\
\|T_5\| &\le B_5 = |\mathcal{E}|\tfrac{\sqrt{N}}{2} + \sqrt{v_s}\,|\mathcal{E}|\,\tfrac{N}{2}\,\tau_t,\\
\|T_6\| &\le B_6 = \big(N(v_u\beta_{\mathrm{pk}}^2 + \sigma_e^2 + \sigma_e^2\beta_s^2)\big)^{1/2} + \big(2t(v_u\gamma_{\mathrm{pk}}^2 + \sigma_e^2 + \sigma_e^2\gamma_s^2)\big)^{1/2},\\
\|T_7\| &\le B_7 = \sqrt{N}/2, \qquad \|T_8\| \le B_8 \quad (\textit{defined in the proof}).
\end{aligned}$$

Here $\beta_{\mathrm{pk}} = \sqrt{2}\sigma_e(\sqrt{N}+\sqrt{2t})$, $\gamma_{\mathrm{pk}} = (4\sigma_e^2 N\ell_N)^{1/2}$, $\beta_s = \sqrt{v_s}(\sqrt{N}+\sqrt{2t})$ *and* $\gamma_s = (2v_s N\ell_N)^{1/2}$.

Proof. Three facts are used throughout.

- By Lemma 1.7, the main part of $\sum_e \Pi_e(y_e)$ has norm $\sqrt{2/N}(\Omega_u/q_1)(\sum_e |\tau(y_e)_{k_e}|^2)^{1/2}$, because the entries occupy distinct slots.
- By Lemma 3.6, the polynomials $\tilde{D}$, rd_1, ε_1 and ε_1' are independent of the random variables they multiply. Corollary 3.3 therefore applies with them treated as fixed, and with no assumption on their distribution. This is the worst case over data required by the review.
- The deterministic roundings ε_0, ε_0' and rd_0 are bounded only through $\|\cdot\|_\infty \le \frac{1}{2}$.

T_1. By Lemma 1.5, $\tau(\mathrm{A}x)_k = \frac{N}{2}(x_0 + ix_{N/2})$. The coefficients $(fg)_0$ and $(fg)_{N/2}$ are inner products of f with signed permutations of g. So $e_R \mapsto (\frac{N}{2}x^R_{e,0}, \frac{N}{2}x^R_{e,N/2})_e$ is linear with Frobenius norm $\frac{N}{2q_2}\sqrt{2}(\sum_{e,h}\|m_{V_e,h}\|^2)^{1/2} \le \frac{N}{2q_2}\sqrt{2}M_V$, by Lemma 3.10. Entries that share R's ciphertext share e_R; the general form of Corollary 3.3 allows this. The coordinates of e_R are σ_e^2-subgaussian (Lemma 3.1).

T_2. The deterministic part satisfies $\frac{N}{2}|\varepsilon_{0,0} + i\varepsilon_{0,N/2}| \le \frac{N}{2}\cdot\frac{1}{\sqrt{2}}$ per entry. The random part is linear in s. Its $2|\mathcal{E}|$ rows have norm at most $\frac{N}{2}\|\varepsilon_{1,e}\| \le \frac{N}{2}\cdot\frac{\sqrt{N}}{2}$.

T_3. Apply Lemma 1.6 with $f = P^{-1}\tilde{D}_{e,i,j}$ and $\mathcal{H}' = \mathcal{H}_{i+1}$. This gives coefficient vectors with $\|\beta\| \le N\|f\|_2 \le N\cdot\sqrt{N}\varsigma_\uparrow\tilde{Q}_j/(2P)$. The coordinates of $e^{\mathrm{rk}} = e_R^{\mathrm{rk}} + e_O^{\mathrm{rk}}$ are $2\sigma_e^2$-subgaussian, and different (i,j) have independent errors. Sum the squared norms over (e,i,j).

T_4. For rd_0, Lemma 1.6 with $f=1$ gives $|\sum_{g\in\mathcal{H}'}\sigma(\mathrm{rd}_0)_{hg}| \le \|\mathrm{rd}_0\|_2\sqrt{N|\mathcal{H}'|} \le \frac{N}{2}\sqrt{|\mathcal{H}_{i+1}|}$. For $\mathrm{rd}_1 s$, take $f = \mathrm{rd}_1$: then $\|\beta\| \le N\|\mathrm{rd}_1\| \le N^{3/2}/2$ for each i, and at most $LN^{3/2}/2$ after summing over i for a fixed entry.

T_5. The deterministic part has $\|\sum_e \varepsilon_{0,e}'\| \le |\mathcal{E}|\sqrt{N}/2$. For the random part, $F = \sum_e \varepsilon_{1,e}'$ is independent of s, with $\|F\|_2 \le |\mathcal{E}|\sqrt{N}/2$ and $\|F\|_\infty^{\mathrm{can}} \le \|F\|_1 \le |\mathcal{E}|N/2$. Apply Corollary 3.3.

T_6. Condition on (e^{pk}, s), which is independent of the fresh (u, e_0, e_1). Apply Corollary 3.3 with $(f_j) = (e^{\mathrm{pk}}, 1, s)$. Then bound $\|e^{\mathrm{pk}}\|, \|e^{\mathrm{pk}}\|_\infty^{\mathrm{can}}, \|s\|, \|s\|_\infty^{\mathrm{can}}$ by Lemma 3.4, which costs four further events.

T_8. By Lemma 1.7, $\|T_8\| \le \frac{N}{2q_1}\sum_e\|Y_e\|$. We have $\|\mathrm{A}y\|_2 = \frac{N}{2}(y_0^2 + y_{N/2}^2)^{1/2}$, which is at most $\sqrt{N/2}(q_1/\Omega_u)(B_1+B_2)$ for each entry, and $\|T_{i+1}w\| \le |\mathcal{H}_{i+1}|\|w\|$. A union bound over the $|\mathcal{E}|L$ key switchings with $t' = t + \ln(|\mathcal{E}|L)$ bounds each $\|e_{\mathrm{ks}}\|$ by $\epsilon_{\mathrm{ks}} = \sqrt{2}\sigma_e d_{\mathrm{num}}\frac{N\tilde{Q}_{\max}}{2P}\tau_{t'} + \frac{\sqrt{N}}{2} + \sqrt{v_s}\frac{N}{2}\tau_{t'}$. This gives

$$B_8 = \tfrac{N}{2q_1}\big(|\mathcal{E}|\sqrt{N/2}\,(q_1/\Omega_u)(B_1+B_2) + |\mathcal{E}|\tfrac{N}{2}\epsilon_{\mathrm{ks}}\big).$$

□

Theorem 3.12 (Obligation O1 discharged)**.** *Under (M1)–(M5), for every dataset D in the clipped domain and every value of the public randomness,*

$$\Pr\big[\|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s} := B_1 + \cdots + B_8\big] \ge 1 - \delta_B,$$

with the probability taken over the key shares, all errors and the fresh randomness only. The centre is $f_s(D)$ of Definition 3.7. The bound covers the whole ring element, in the coefficient ℓ_2 norm.

Proof. Combine Proposition 3.8, Proposition 3.11, the triangle inequality and a union bound over the n_{ev} events. □

Corollary 3.13 (Sensitivity of the centre)**.** *Let $U_{\max,e}$ bound $|U_e|$: 1 for $\mathbf{b}$, and B_R for $\dot{\mathbf{x}}_j$. Put $\nu_U = (\sum_e U_{\max,e}^2)^{1/2}$ and $\nu_V = \sup_{D\sim D'}(\sum_e (V_e - V_e')_t^2)^{1/2} \le 2\Gamma_{\mathcal{E}}^{1/2}$. Then, with $\Omega_U = \Omega_V = \Omega_u = q_1 = q_2$,*

$$\sup_{D\sim D'} \|f_s(D) - f_s(D')\|_2 \le \widehat{\Delta}_{\mathrm{enc},s} := \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N},$$

$$\Delta_{x,s} = \sqrt{N/2}\,(\nu_V + \sqrt{N}\nu_U) + \tfrac{\sqrt{N}}{2}\sqrt{|\mathcal{E}|}\,\Delta_s + \tfrac{N^{3/2}}{4q_1}\sqrt{|\mathcal{E}|}(\nu_V + \sqrt{N}\nu_U).$$

Terms of order $N^{3/2}\sqrt{|\mathcal{E}|}/\Omega$, which come from the $\sqrt{N}/2$ parts of $\|m_U\|$, are omitted; they are below 2^{-20} for the example and are covered by adding 1 to $\Delta_{x,s}$. For the example, $\Delta_{x,2} \approx 2^{18.8}$ against $\sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_2 \approx 2^{52.4}$. So $\widehat{\Delta}_{\mathrm{enc}}$ exceeds the specification's Δ_{enc} by a relative $2^{-33.7}$, and ρ_s by a relative $2^{-32.7}$.

Proof. Neighbours differ in one row, which sits in chunk h^* and slot l^*. Let $\Delta m_e = \mathsf{Ecd}(V_e[h^*]) - \mathsf{Ecd}(V_e'[h^*]) = \Omega_V\tau^{-1}(\Delta V_e \mathbf{u}_{l^*}) + \delta r_e$, with $\|\delta r_e\|_\infty \le 1$.

The slot value. By Lemma 1.5, $\mathrm{A}(m_{U_e,h^*}\Delta m_e)$ has all slots equal to

$$c_e = \sum_l \tau(m_U)_l \tau(\Delta m_e)_l = \Omega_U\Omega_V U_{e,l^*}\Delta V_e + \Omega_V\tau(r_U)_{l^*}\Delta V_e + \sum_l \tau(m_U)_l\tau(\delta r_e)_l.$$

The first term is $\Omega_U\Omega_V$ times the change of entry e, and these changes have ℓ_2 norm at most Δ_s. Bound $|\tau(r_U)_{l^*}| \le \|r_U\|_1 \le N/2$. By Cauchy–Schwarz and Lemma 1.1, the last sum is at most $\frac{N}{2}\|m_U\|\|\delta r\| \le \frac{N}{2}(\Omega_U U_{\max} + \frac{\sqrt{N}}{2})\sqrt{N}$.

Masking. Apply Lemma 1.7. The main part contributes $\sqrt{2/N}\,(\Omega_u/(q_1q_2))\|(c_e)_e\|$, and the spill contributes $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e|$.

Rounding. Finally, $\|\lfloor x\rceil - \lfloor x'\rceil\| \le \|x - x'\| + \sqrt{N}$. □

3.5 Numerical evaluation

Take the example parameters: $N = 2^{14}$, $L = 13$, $d_R = 4$, $d_O = 6$, $\bar{n}_O = 10^6$ (so $m = 123$), $B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $\sigma_e = 3.2$, ternary s_R, s_O, u, $\delta_B = 2^{-50}$ (so $t = 37.1$ and $\tau_t = 9.62$), $d_{\mathrm{num}} = 2$ and $\varsigma_\uparrow = 1$. Then $|\mathcal{E}_G| = 51$, $|\mathcal{E}_2| = 63$ (joint mode), $\Gamma_{\mathcal{E}_G} = 1711$ and $\Gamma_{\mathcal{E}_2} = 2147$.

The budget is $\varepsilon = 1$ and $\delta_{\mathrm{run}} = 10^{-9}$, with three releases, $\kappa = 40$ and $f_1 = 0.1$. The conversion of Proposition 5.4 gives $\rho_{\mathrm{run}} = 0.01496$ and a stage-2 slot s.d. of 525.

The column $\log_2\Omega$ is the least scale meeting CALIBRATE's condition $\sigma_{c,s} \ge 2^\kappa B_{\mathrm{tot},s}/\sqrt{2\pi}$, with $\sigma_{c,s} = \Delta_{\mathrm{enc},s}/\sqrt{2\rho_s}$. The condition is binding for stage 2. Script: `scratch/btot.py`.

Configuration (stage 2)	B_1	B_2	B_3	B_4	B_5	B_6	$\log_2 B_{\text{tot}}$	$\log_2 \Omega$
(A) Algorithm 4 as specified, $\tilde{Q}_{\max}/P = 1$	20.9	19.5	24.3	23.7	22.5	17.2	25.34	61.5
(A′) as (A), $P \geq 2^{20}\tilde{Q}_{\max}$	20.9	19.5	4.3	23.7	22.5	17.2	24.39	60.5
(C) EvalSum before the first rescale; stochastic rounding; one final rescale; $P \geq 2^{10}\tilde{Q}_{\max}$	20.9	14.1	< 0	< 0	14.0	17.2	21.07	57.2
(C′) as (C), with $\Omega_U = q_2 = 2^{60}$	13.4	14.1	< 0	< 0	14.0	17.2	17.56	53.7
Fresh-encryption error only (what v3.1 assumed)	–	–	–	–	–	17.2	17.17	53.3

Table 1: $\log_2$ of the bounds of Proposition 3.11 for stage 2 of the example. In all rows $B_7 = 2^6$ and $B_8 < 1$. For stage 1 the required scale is 1.4 to 1.6 bits lower; for (A) it is 59.9. Using Lemma 2.3(a) instead of (b) adds 0.33 bits.

Reading the table.

- The specification's $\log_2 \Omega \approx 52.5$ floods only the fresh-encryption error, measured by its *mean* ℓ_2 norm of about $2^{16.4}$. A high-probability bound on that term alone already needs 53.3.
- For the circuit exactly as specified, the worst-case analysis is dominated by three terms:
  - the gadget term B_3, unless $P \gg \tilde{Q}_{\max}$;
  - the ModDown roundings B_4;
  - the $|\mathcal{E}|$ separate second rescales B_5.

  The latter two are large because roundings are treated as adversarially aligned with a slot. That is the price of a bound with no randomness assumption on rounding errors.
- Variant (C) removes these terms in three ways:
  - EvalSum at level 2, so that key-switching errors are divided by q_2 and the first-rescale rounding is not traced;
  - stochastic rounding, so that rounding errors are mean-zero and $\frac{1}{4}$-subgaussian given the ciphertext, by Lemma 3.1(c);
  - a single rescale after summing the masked products.

  (C′) also shrinks T_1 by encoding R's vectors at a larger scale.

The bounds for (C) and (C′) use the same lemmas. We give the formulas in the script rather than repeating the proofs. [**Not fully rigorous:** The proofs for variants (C) and (C′) are only sketched. The T_2 and T_5 formulas for (C) apply Corollary 3.3 conditionally on s, to the independent stochastic-rounding coins.]

3.6 Empirical sanity check with OpenFHE

Setup. We ran the circuit shape of Algorithm 4 in OpenFHE [1], through its Python bindings, (`scratch/ofhe_check.py`). The run used:

- a single key, FIXEDMANUAL, $N = 2^{14}$, scale 2^{50}, depth 2;
- one entry, with $m \in \{4, 16\}$ chunks;
- 0/1 vectors U and $|V| \leq o/2$.

Two runs on the same data with independent keys and randomness isolate the randomness-dependent error, because the deterministic part cancels in the difference.

Results.

- The measured coefficient ℓ_2 norm of the random error was $2^{14.0}$ for $m = 4$ and $2^{15.7}$ for $m = 16$. Theorem 3.12 at the same setting gives $2^{21.8}$ and $2^{21.9}$. The bound holds, with a margin of 6

to 8 bits, as expected for a worst-case, high-probability bound.

- The deviation of the decryption from $\mathsf{Ecd}(\mathbf{z})$ was $2^{23.0}$. It is dominated by a deterministic relative error of $8.7 \cdot 10^{-10} \approx 2^{-30}$ at the target slot, from decoding with the nominal scale 2^{50} rather than the exact Ω_{fin}. This confirms Remark 3.9 and Correction 2.

4 Decoupled scales, OpenFHE's key switching and the decoding offset

The evaluation runs circuit (A) with *decoupled* encoding scales ($\Omega_U = 2^{68}$, $\Omega_u = q_1$, Ω_V chosen by CALIBRATE) and with the hybrid key switching of OpenFHE 1.5.1. Proposition 3.11 and Theorem 3.12 are already stated for general $\Omega_U, \Omega_V, \Omega_u, q_1, q_2$: no step of their proofs uses $\Omega_U = q_2$ or $\Omega_V = \Omega_u = q_1$. Three things are not covered by Sections 3.2–3.5: OpenFHE's uncentred gadget (Lemma 4.1), the sensitivity of the centre with decoupled scales (Lemma 4.3; Corollary 3.13 assumes equal scales), and a public bound on the decoding offset (Proposition 4.5), which the statistical part needs. This section proves them. The proofs were first written in the companion book on the protocol (its Appendix B, Lemmas `btot.gadget` and `btot.sensZ`, Proposition `btot.offset`, and Chapter 9, Corollary `agg.sensZ` and Lemma `agg.offset`); they were checked here step by step, with the corrections noted, and the numbers recomputed (script `experiments/proofs_check/check_btot_offset.py` of the repository).

4.1 OpenFHE's uncentred key switching

(M3′) As (M3), with three differences. (i) The digit $\tilde{D}_j(c)$ is the integer polynomial $\sum_{i\in j}[c_i(\tilde{Q}_j/q_i)^{-1}]_{q_i}(\tilde{Q}_j/q_i)$, with residues $[\cdot]_{q_i} \in [0, q_i)$; so $\tilde{D}_j(c) \equiv c \pmod{\tilde{Q}_j}$ and $\tilde{D}_j(c) \in [0, k_j\tilde{Q}_j)^N$ (fast base extension without centring). (ii) ModDown is $\mathrm{MD}(x) = (x - y_P)/P$, with $y_P = \sum_i [x_i(P/p_i)^{-1}]_{p_i}(P/p_i) \in [0, k_P P)^N$ and $y_P \equiv x \pmod{P}$, where $p_1, \ldots, p_{k_P}$ are the special primes; so $\mathrm{MD}(x) = x/P + \mathrm{rd}$ with $\mathrm{rd} = -y_P/P \in (-k_P, 0]^N$, the division being exact. (iii) The key switch from s to $\psi_g^{-1}(s)$ is applied *before* the automorphism: the key is $b_j = -a_j\psi_g^{-1}(s) + P\mathsf{g}_j s + e_j^{\mathrm{rk}}$, and the rotation outputs $(\psi_g(c_0 + \mathrm{MD}(\sum_j \tilde{D}_j(c_1)b_j)),\ \psi_g(\mathrm{MD}(\sum_j \tilde{D}_j(c_1)a_j)))$.

In the notation of Proposition 3.11, (M3′) gives $\|\tilde{D}_j\|_\infty < k_j\tilde{Q}_j \le \varsigma_\uparrow \tilde{Q}_{\max}/2$ with $\varsigma_\uparrow = 2\max_j k_j$, and $\|\mathrm{rd}\|_\infty < k_P = \varsigma_\downarrow/2$ with $\varsigma_\downarrow = 2k_P$, against $\varsigma_\uparrow \le \max_j k_j$ and $\varsigma_\downarrow = 1$ for the centred gadget of (M3).

Lemma 4.1 (Uncentred digits and ModDown rounding)**.** *Assume (M1), (M2), (M3′), (M4), (M5). Then Proposition 3.8 holds verbatim with every $\tilde{D}_{e,i,j}$, $e_j^{\mathrm{rk},(i)}$, $\mathrm{rd}_{0,e,i}$, $\mathrm{rd}_{1,e,i}$ replaced by its image under the automorphism ψ_{g_i} of rotation i. Proposition 3.11 and Theorem 3.12 hold with $\varsigma_\uparrow = 2\max_j k_j$ in B_3 and ϵ_{ks}, and with the ModDown terms (all of B_4, and the terms $\sqrt{N}/2$ and $\sqrt{v_s}\frac{N}{2}\tau_{t'}$ of ϵ_{ks}) multiplied by $\varsigma_\downarrow = 2k_P$. B_1, B_2, B_5, B_6, B_7 are unchanged. No mean-zero or symmetry property of the digits or of the roundings is used.*

Proof. The switch. Let $c_0 + c_1 s \equiv \mu \pmod{Q_1}$. Modulo PQ_1, $\sum_j \tilde{D}_j(c_1)(b_j + a_j\psi_g^{-1}(s)) = P\sum_j \tilde{D}_j(c_1)\mathsf{g}_j s + \sum_j \tilde{D}_j(c_1)e_j^{\mathrm{rk}}$, and $\sum_j \tilde{D}_j(c_1)\mathsf{g}_j \equiv c_1 \pmod{Q_1}$ by the CRT, because $\tilde{D}_j(c_1) \equiv c_1 \pmod{\tilde{Q}_j}$ holds for uncentred digits as well. $\mathrm{MD}(x) \bmod Q_1$ depends only on $x \bmod PQ_1$ (adding PQ_1K to x adds Q_1K to $\mathrm{MD}(x)$). Dividing by P and adding the roundings, the switched pair

(c'_0, c'_1) satisfies

$$c'_0 + c'_1\psi_g^{-1}(s) \equiv \mu + e'_{\mathrm{ks}} \pmod{Q_1}, \qquad e'_{\mathrm{ks}} = P^{-1}\sum_j \tilde{D}_j(c_1)e_j^{\mathrm{rk}} + \mathrm{rd}_0 + \mathrm{rd}_1\psi_g^{-1}(s).$$

The automorphism afterwards. $\psi_g(c'_0) + \psi_g(c'_1)s = \psi_g(c'_0 + c'_1\psi_g^{-1}(s))$, so the rotated pair decrypts to $\psi_g(\mu) + \psi_g(e'_{\mathrm{ks}})$ under s, and

$$\psi_g(e'_{\mathrm{ks}}) = P^{-1}\sum_j \tilde{D}_j^\star e_j^\star + \mathrm{rd}_0^\star + \mathrm{rd}_1^\star s, \qquad \tilde{D}_j^\star = \psi_g(\tilde{D}_j(c_1)),\ e_j^\star = \psi_g(e_j^{\mathrm{rk}}),\ \mathrm{rd}_b^\star = \psi_g(\mathrm{rd}_b).$$

This is the form of Lemma 3.5. The proof of Proposition 3.8 only adds such errors and applies the traces T_{i+1} to them, so the decomposition holds with the starred quantities.

The hypotheses of the bounds. (a) ψ_g is a signed permutation of coefficients (Lemma 1.4), so the starred digits and roundings have the same ℓ_∞, ℓ_2 and ℓ_1 norms as the unstarred ones. (b) The coordinates of $e_j^\star$ are those of e_j^{rk} up to order and sign; a sign flip preserves v-subgaussianity, so $e_j^\star$ has independent $2\sigma_e^2$-subgaussian coordinates, independent across (i, j). (c) $\tilde{D}_j^\star$ and $\mathrm{rd}_1^\star$ are functions of the c_1-component, which by the induction of Lemma 3.6 (the new step outputs $\psi_g(\mathrm{MD}(\sum_j \tilde{D}_j(c_1)a_j))$ and uses neither c_0, nor a secret, nor an error) is a function of the public randomness and the data only. (d) The rescales are those of (M1). The proofs of B_3, B_4 and B_8 use only $\|\tilde{D}\|_\infty$, $\|\mathrm{rd}\|_\infty$, the proxy of the key errors, (c), and Lemma 1.6, which holds for every polynomial f; so they hold with the stated factors. □

Remark 4.2 (Conformance of OpenFHE 1.5.1 to (M3′); a code reading, not a proof)**.** We re-read the relevant code at tag `v1.5.1` (the headers installed with the implementation are identical to the tagged ones):

- `core/include/lattice/hal/default/dcrtpoly-impl.h`, `ApproxSwitchCRTBasis`: residues are stored in $[0, q_i)$; both code paths compute $\sum_i [x_i\hat{Q}_i^{-1}]_{q_i}[\hat{Q}_i]_p \bmod p$, the residue of the integer $\sum_i [x_i\hat{Q}_i^{-1}]_{q_i}\hat{Q}_i \in [0, k\tilde{Q})$. `ApproxModDown`: `ans` $= (x - \mathrm{partPSwitchedToQ}) \cdot P^{-1}$, with no centring; the t-multiplications are skipped for CKKS, whose noise scale is 1 (`gen-cryptocontext-ckksrns-internal.h`).
- `pke/lib/keyswitch/keyswitch-hybrid.cpp`: `KeySwitchGenInternal` sets $b = -a\,s_{\mathrm{new}} + e + [P]_{q_i}s_{\mathrm{old}}$ on the primes of digit j and $-a\,s_{\mathrm{new}} + e$ elsewhere, which is $P\mathbf{g}_j s_{\mathrm{old}}$; `EvalKeySwitchPrecomputeCore` keeps the digit's own residues and extends them by `ApproxSwitchCRTBasis`; `EvalFastKeySwitchCore` applies `ApproxModDown` to both components; `KeySwitchInPlace` adds the first to c_0 and replaces c_1 by the second.
- `pke/lib/schemebase/base-leveledshe.cpp`, `EvalAutomorphism`: `KeySwitchInPlace`, then `AutomorphismTransform` of both components; CKKS does not override it. `base-multiparty.cpp`, `MultiEvalAutomorphismKeyGen`: the new key is $\psi_{g^{-1}}(s_P)$.
- The implementation (`implementation/native/vfr_native.cpp`) builds the rotation-key shares in this layout itself (`rotkey_share`: $b_j = -a_j\psi_g^{-1}(s) + e + P\mathbf{g}_j s$), sums them, calls the scheme's `EvalAutomorphism` inside its own EvalSum, and uses its own centred deterministic rescale (`rescale_poly`, remainder in $(-q/2, q/2]$), its own encodings and its own release. So (M1) holds as stated and only key switching is OpenFHE's.

These are statements about a program and are not proved. (M5) for OpenFHE's table sampler, which draws the errors, remains a numerical check (Section 3.6; the paper's Check (T4)).

4.2 The sensitivity of the centre with decoupled scales

For release s let $\nu_U = (\sum_{e\in\mathcal{E}} U_{\max,e}^2)^{1/2}$, with $U_{\max,e} = 1$ for $\mathbf{b}$ and B_R for a column of $\dot{\mathbf{x}}$, and $\nu_V = \sup_{D\sim D'} \max_t (\sum_{e\in\mathcal{E}} (V_{e,t} - V'_{e,t})^2)^{1/2} \le 2\Gamma_{\mathcal{E}}^{1/2}$ (Lemma 3.10).

Lemma 4.3 (Sensitivity of the centre, decoupled scales)**.** *Assume (M1), (M2) with arbitrary $\Omega_U, \Omega_V, \Omega_u, q_1, q_2$, and that* Ecd *rounds to the nearest integer coefficient-wise. Put*

$$\begin{aligned}
\epsilon_x &= \frac{\Omega_u}{q_1 q_2}\,\frac{N^{3/2}\sqrt{|\mathcal{E}|}}{2\sqrt{2}} + \frac{N^{5/2}|\mathcal{E}|}{8 q_1 q_2},\\
\Delta_{x,s} &= \sqrt{\tfrac{N}{2}}\,\frac{\Omega_u}{q_1 q_2}\big(\Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_U\big) + \frac{\sqrt{N}}{2}\,\frac{\Omega_U \Omega_V}{q_1 q_2}\sqrt{|\mathcal{E}|}\,\Delta_s\\
&\quad + \frac{N^{3/2}}{4 q_1 q_2}\sqrt{|\mathcal{E}|}\big(\Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_U\big) + 1.
\end{aligned}$$

For every pair of neighbours $D \sim D'$ (differing in one row of O*, matched or not) and every input and alignment of* R*, $\|\widehat{Z}_s(D) - \widehat{Z}_s(D')\|_2 \le \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} - 1 + \epsilon_x$; hence, if $\epsilon_x \le 1$,*

$$\|f_s(D) - f_s(D')\|_2 \le \widehat{\Delta}_{\mathrm{enc},s} := \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N}.$$

With equal scales this reduces to Corollary 3.13.

Proof. The changed row sits in chunk h^* and slot l^*; only the encodings m_{V_e,h^*} change. Write $\Delta V_e = V_{e,l^*} - V'_{e,l^*}$ and $\Delta m_e = m_{V_e,h^*} - m'_{V_e,h^*} = \Omega_V \tau^{-1}(\Delta V_e \mathbf{u}_{l^*}) + \delta r_e$ with $\|\delta r_e\|_\infty \le 1$. Then $\widehat{Z}_s(D) - \widehat{Z}_s(D') = \sum_e (q_1 q_2)^{-1} y_e m_{u_e}$ with $y_e = \mathrm{A}(m_{U_e,h^*}\Delta m_e)$.

Slot value. By Lemma 1.5 and multiplicativity of τ, every slot of y_e equals $c_e = \sum_l \tau(m_{U_e,h^*})_l \tau(\Delta m_e)_l$. With $m_{U_e,h^*} = \Omega_U \tau^{-1}(U_e[h^*]) + r_{U_e}$,

$$c_e = \Omega_U\Omega_V\, U_{e,l^*}\Delta V_e + \Omega_V\, \tau(r_{U_e})_{l^*}\Delta V_e + \sum_l \tau(m_{U_e,h^*})_l \tau(\delta r_e)_l.$$

The vector $(U_{e,l^*}\Delta V_e)_e$ is the change of the statistic, of norm at most Δ_s; it is zero if the row is unmatched, because R's vectors vanish there. $|\tau(r_{U_e})_{l^*}| \le \|r_{U_e}\|_1 \le N/2$, so the second terms have norm at most $\Omega_V \frac{N}{2}\nu_V$. By Cauchy–Schwarz and Lemma 1.1 the third term is at most $\frac{N}{2}\|m_{U_e,h^*}\|\|\delta r_e\| \le \frac{N}{2}(\Omega_U U_{\max,e} + \frac{\sqrt{N}}{2})\sqrt{N}$, using $\|m_U\| \le \Omega_U\sqrt{2/N}\|U_e[h^*]\| + \sqrt{N}/2$ and $\|U_e[h^*]\| \le \sqrt{N/2}\,U_{\max,e}$. So

$$\|(c_e)_e\| \le \Omega_U\Omega_V\Delta_s + \tfrac{N}{2}\big(\Omega_V\nu_V + \sqrt{N}\,\Omega_U\nu_U\big) + \tfrac{N^2}{4}\sqrt{|\mathcal{E}|}.$$

Masking. Split $m_{u_e} = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$. The main part has norm $\sqrt{2/N}\,(\Omega_u/(q_1q_2))\|(c_e)_e\|$ (Lemma 1.7). For the spill, y_e has only the coefficients of 1 and $X^{N/2}$ (Lemma 1.5): $y_e = a + bX^{N/2}$ with $a^2 + b^2 = |c_e|^2$. Multiplication by $X^{N/2}$ is an orthogonal map J with $J^\top = -J$, so $\langle r, Jr\rangle = 0$ and $\|y_e r_{u_e}\|^2 = |c_e|^2\|r_{u_e}\|^2 \le |c_e|^2 N/4$. Hence the spill is at most $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e| \le \frac{\sqrt{N}}{2q_1q_2}\sqrt{|\mathcal{E}|}\|(c_e)_e\|$.

Collecting. With $\Omega_U\Omega_V\Omega_u/(q_1q_2) = \Omega_{\mathrm{fin}}$, the main part gives $\sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s$, the first term of $\Delta_{x,s}$, and $\sqrt{2/N}(\Omega_u/(q_1q_2))\frac{N^2}{4}\sqrt{|\mathcal{E}|}$; the spill gives the second and third terms and $\frac{\sqrt{N}}{2q_1q_2}\frac{N^2}{4}|\mathcal{E}|$. The two leftovers sum to ϵ_x. Finally $\|\lfloor x\rceil - \lfloor x'\rceil\| \le \|x - x'\| + \sqrt{N}$. □

Remark 4.4 (Second reading)**.** The book states the lemma for a changed row that is matched; the proof covers any row, as shown (an unmatched row contributes ΔV_e only through the rounding

cross terms), and the paper's adjacency (Definition `def:adj`) changes a row "matched or not". A brute-force check at $N = 16$ with decoupled scales $(\Omega_U, \Omega_V, \Omega_u, q_1, q_2) = (2^{10}, 2^9, 2^{10}, 2^{10}, 2^9)$ and exact rational arithmetic gives a largest ratio 0.93 of the change of f_s to the bound over 30 random instances. If the encoder's output differs from exact rounding by at most δ per coefficient, every rounding bound $\frac{1}{2}$ becomes $\frac{1}{2} + \delta$ and the lemma holds with the first three terms of $\Delta_{x,s}$ multiplied by $(1 + 2\delta)^2$; the implementation's double-double encoder is not analysed here.

4.3 The decoding offset

Let $\zeta_{\mathcal{E}}^2 = \sup_{\text{rows}} \sum_e (U_{e,t} V_{e,t})^2$, the largest squared contribution of one row to the statistic: with $r = B_R^2$, $\zeta_{\mathcal{E}_G}^2 \le o^2 + ro$ and $\zeta_{\mathcal{E}_2}^2 \le o^2 + r(o + w) + ow + w^2$ (joint mode), by the block computation of Lemma 3.10. Let $n_{\max}$ bound the number of matched rows ($n_R^{\max}$ in the protocol).

Proposition 4.5 (A public bound on the decoding offset)**.** *Assume (M1), (M2) and exact rounding in* Dcd*. Put*

$$A_s = \frac{N\sqrt{m}}{2\sqrt{2}} \Big(\frac{\sqrt{n_{\max}}\, \nu_U}{\Omega_V} + \frac{(\bar{n}_O \Gamma_{\mathcal{E}})^{1/2}}{\Omega_U} \Big) + \frac{\sqrt{|\mathcal{E}|}\, mN^2}{8\, \Omega_U \Omega_V}.$$

For every dataset in the clipped domain with at most $n_{\max}$ matched rows, the decoded centre satisfies, on the slots of the entries,

$$\Big\| \big(\mathsf{Dcd}(f_s(D))_{k_e} - \mathbf{z}_e(D) \big)_{e \in \mathcal{E}} \Big\|_2 \le \bar{d}_s, \qquad \bar{d}_s := A_s + \frac{N\sqrt{|\mathcal{E}|}}{2\sqrt{2}\, \Omega_u} \big(n_{\max} \zeta_{\mathcal{E}} + A_s \big) + \frac{N}{2\sqrt{2}\, \Omega_{\text{fin}}},$$

with $\mathsf{Dcd}(v) = \tau(v)/\Omega_{\text{fin}}$*, and, in the coefficient domain,*

$$\|\widehat{Z}_s(D) - \Omega_{\text{fin}} \tau^{-1}(\mathbf{z}_s(D))\|_2 \le d_{\max},$$

$$d_{\max} := \sqrt{\tfrac{2}{N}}\, \frac{\Omega_u}{q_1 q_2}\, \bar{r}_s + \frac{\sqrt{N}}{2 q_1 q_2} \sqrt{|\mathcal{E}|} \big(\Omega_U \Omega_V n_{\max} \zeta_{\mathcal{E}} + \bar{r}_s \big), \qquad \bar{r}_s = \Omega_U \Omega_V A_s.$$

Proof. Write $m_{U_e,h} = \Omega_U \tau^{-1}(U_e[h]) + r_{U,e,h}$ and $m_{V_e,h} = \Omega_V \tau^{-1}(V_e[h]) + r_{V,e,h}$, with roundings of sup-norm at most $\frac{1}{2}$, so $\|\tau(r)\| = \sqrt{N/2}\|r\| \le N/(2\sqrt{2})$. As in the previous proof, every slot of $y_e = \mathrm{A}(\sum_h m_{U_e,h} m_{V_e,h})$ equals

$$c_e = \Omega_U \Omega_V \mathbf{z}_e + \bar{r}_e, \qquad \bar{r}_e = \Omega_U a_e + \Omega_V b_e + d_e,$$

with $a_e = \sum_h \langle U_e[h], \tau(r_{V,e,h}) \rangle$, $b_e = \sum_h \langle \tau(r_{U,e,h}), V_e[h] \rangle$, $d_e = \sum_h \langle \tau(r_{U,e,h}), \tau(r_{V,e,h}) \rangle$ (bilinear sums over the slots). By Cauchy–Schwarz over slots and chunks, $|a_e| \le \frac{N}{2\sqrt{2}} \sqrt{m} \|U_e\|$, $|b_e| \le \frac{N}{2\sqrt{2}} \sqrt{m} \|V_e\|$, $|d_e| \le mN^2/8$. R's vectors vanish off the matched rows, so $\sum_e \|U_e\|^2 \le n_{\max} \nu_U^2$, and $\sum_e \|V_e\|^2 \le \bar{n}_O \Gamma_{\mathcal{E}}$ (Lemma 3.10); hence $\|(\bar{r}_e)_e\| \le \Omega_U \Omega_V A_s = \bar{r}_s$.

Slots. Since $\tau(y_{e'} r_{u_{e'}}) = c_{e'} \tau(r_{u_{e'}})$, $\tau(\widehat{Z}_s)_k = (q_1 q_2)^{-1} \sum_{e'} c_{e'} (\Omega_u [k = k_{e'}] + \tau(r_{u_{e'}})_k)$. Dividing by Ω_{fin},

$$\mathsf{Dcd}(\widehat{Z}_s)_{k_e} - \mathbf{z}_e = \frac{\bar{r}_e}{\Omega_U \Omega_V} + \sum_{e'} \frac{c_{e'}\, \tau(r_{u_{e'}})_{k_e}}{\Omega_U \Omega_V \Omega_u}.$$

The first term has norm at most A_s over the entries. For the second, $\|(\tau(r_{u_{e'}})_{k_e})_e\| \le \|\tau(r_{u_{e'}})\| \le N/(2\sqrt{2})$, so it is at most

$$\tfrac{N}{2\sqrt{2}\Omega_u} \textstyle\sum_{e'} |c_{e'}|/(\Omega_U \Omega_V) \le \tfrac{N\sqrt{|\mathcal{E}|}}{2\sqrt{2}\Omega_u} (\|\mathbf{z}\| + A_s),$$

and $\|\mathbf{z}\| \le n_{\max}\zeta_{\mathcal{E}}$ by the triangle inequality over the matched rows. Finally $f_s - \widehat{Z}_s$ has sup-norm at most $\frac{1}{2}$, so its slot vector has norm at most $N/(2\sqrt{2})$, which contributes $N/(2\sqrt{2}\,\Omega_{\mathrm{fin}})$. Taking real parts does not increase norms.

Coefficients. The main part of $\widehat{Z}_s$ is

$$\textstyle\sum_e (\Omega_u/(q_1q_2))y_e\tau^{-1}(\mathbf{u}_{k_e}) = \Omega_{\mathrm{fin}}\tau^{-1}(\mathbf{z}_s) + (\Omega_u/(q_1q_2))\tau^{-1}(\sum_e \bar{r}_e\mathbf{u}_{k_e}),$$

whose second piece has norm $\sqrt{2/N}(\Omega_u/(q_1q_2))\|(\bar{r}_e)\|$ (Lemma 1.2(a)). The spill is at most $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e|$, as in the previous proof. □

The numbers at the evaluation setting. Chain $[60, 59, 59, 59]$ with $q_0 = p_0p_1 \approx 2^{119}$, $q_1, q_2 \approx 2^{59}$, two special primes of 60 bits, $d_{\mathrm{num}} = 2$ digits at the EvalSum level ($\max_j k_j = 2$, $\tilde{Q}_{\max}/P \approx 2^{-1}$), $\Omega_U = 2^{68}$, $\Omega_u = q_1$, simulation population ($d_R = 4$, $d_O = 6$, $(B_R, B_O, B_y) = (5.59, 6.24, 2.89)$), $\varepsilon = 1$, $\delta = 10^{-9}$, $\kappa = 40$, previous setting ($f_1 = 0.1$), recomputed with the implementation's Calibrate:

- $\log_2 \Omega_{\mathrm{fin}} = 62.75$; $B_{\mathrm{tot},1} = 2^{26.43}$ and $B_{\mathrm{tot},2} = 2^{26.59}$ by Lemma 4.1 ($\varsigma_\uparrow = 4$, $\varsigma_\downarrow = 4$), the same at $\bar{n}_O = 4{,}096$, 16,384 and 10^6;
- $\Delta_{x,2} = 2^{27.57}$ and $\Delta_{x,1} = 2^{27.46}$; $\epsilon_x = 2^{-36.5}$ and $2^{-36.7}$, so the "+1" of $\Delta_{x,s}$ covers it;
- $\bar{d}_2 = 9.9 \cdot 10^{-9}$ at $\bar{n}_O = 4{,}096$ with $n_{\max} = n_R^{\max} = 2{,}000$; $2.0 \cdot 10^{-8}$ with $n_{\max} = \bar{n}_O = 4{,}096$; $1.6 \cdot 10^{-8}$ and $4.8 \cdot 10^{-6}$ at $\bar{n}_O = 10^6$ with $n_{\max} = 2{,}000$ and 10^6; against $\sigma_2 = 525$;
- $d_{\max} = 2^{30.3}$ at $\bar{n}_O = 10^6$ with $n_{\max} = 2{,}000$, and $2^{38.6}$ with $n_{\max} = 10^6$.

A brute-force check at $N = 16$ with decoupled scales gives a largest ratio 0.07 of the offset to $\bar{d}_s$.

5 Output privacy (Theorem 1)

5.1 Differential-privacy tools

Definition 5.1 (zCDP, Bun and Steinke [3])**.** A randomized mechanism $\mathcal{M}$ is ρ-zCDP with respect to $\sim$ if $D_\alpha(\mathcal{M}(D)\|\mathcal{M}(D')) \le \rho\alpha$ for all $D \sim D'$ and all $\alpha > 1$. Here D_α is the Rényi divergence of order α.

Lemma 5.2 ([3], composition and post-processing)**.**

- Composition. *Suppose $\mathcal{M}_1$ is ρ_1-zCDP, and $\mathcal{M}_2(\cdot, y)$ is ρ_2-zCDP for every y. Then $D \mapsto (y, \mathcal{M}_2(D, y))$ with $y \leftarrow \mathcal{M}_1(D)$ is $(\rho_1 + \rho_2)$-zCDP.*
- Post-processing. *$g \circ \mathcal{M}$ is ρ-zCDP for every randomized g that does not depend on D.*

Proposition 5.3 (The ideal mechanism; multivariate discrete Gaussian)**.** *Let $f : \mathcal{D} \to \mathbb{Z}^N$ satisfy $\sup_{D\sim D'}\|f(D) - f(D')\|_2 \le \widehat{\Delta}$, and let $\mathcal{M}(D) = f(D) + \eta$ with $\eta \sim D_{\mathbb{Z}^N,\sigma}$. Then $\mathcal{M}$ is $\widehat{\Delta}^2/(2\sigma^2)$-zCDP.*

Hence $\mathcal{M}^{\mathrm{id}}(D) = (f_s(D) + \eta_s)_s$, with independent $\eta_s \sim D_{\mathbb{Z}^N,\sigma_{c,s}}$ and f_s from Definition 3.7, is $\sum_s \rho_s$-zCDP, with $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$.

Proof. Single release. $\mathcal{M}(D)$ and $\mathcal{M}(D')$ are integer shifts of one another, by $\mathbf{e} = f(D) - f(D') \in \mathbb{Z}^N$. Lemma 2.2 gives $D_\alpha \le \alpha\|\mathbf{e}\|^2/(2\sigma^2)$. This is the multivariate statement of [4] [*citation detail unverified:* numbered Theorem 14 in the arXiv version], proved here in a self-contained way.

Several releases. The releases are computed from fixed ciphertexts M_1, M_2, M_y. R's choices between releases do not change what O releases, so non-adaptive composition (Lemma 5.2) applies.

The centre. Corollary 3.13 bounds the sensitivity of f_s. The specification's $\Delta_{\mathrm{enc},s}$ is the sensitivity of $\mathsf{Ecd}_\Omega(\mathbf{z}_s)$ (Lemma 1.2); it must be replaced by $\widehat{\Delta}_{\mathrm{enc},s}$, a relative change of 2^{-33}. □

Proposition 5.4 (Conversion from zCDP, Canonne, Kamath and Steinke [4])**.** *If $\mathcal{M}$ is ρ-zCDP, then it is $(\varepsilon, \delta(\rho,\varepsilon))$-DP for every $\varepsilon \geq 0$, where*

$$\delta(\rho,\varepsilon) = \inf_{\alpha>1} \frac{e^{(\alpha-1)(\alpha\rho-\varepsilon)}}{\alpha-1}\Big(1-\frac{1}{\alpha}\Big)^{\alpha}.$$

The map $\rho \mapsto \delta(\rho,\varepsilon)$ is continuous and nondecreasing, and $\delta(\rho,\varepsilon) \to 0$ as $\rho \to 0$. Hence $\rho_{\mathrm{run}} = \max\{\rho : \delta(\rho,\varepsilon_{\mathrm{run}}) \leq \delta_{\mathrm{DP}}\}$ in CALIBRATE *is well defined, and the ρ_{run}-zCDP mechanism is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP.*

Proof. Let $P = \mathcal{M}(D)$ and $Q = \mathcal{M}(D')$, with $D \sim D'$.

Hockey-stick form. (ε,δ)-indistinguishability is equivalent to $\mathbb{E}_Q[(P/Q - e^{\varepsilon})_+] \leq \delta$ (with the usual convention when $P \not\ll Q$; here both have full support on $\mathbb{Z}^N$).

A pointwise bound. For $x > 0$ and $\alpha > 1$, $(x - e^{\varepsilon})_+ \leq x^{\alpha} c_{\alpha}$, where

$$c_{\alpha} = \sup_{y>0} \frac{(y-e^{\varepsilon})_+}{y^{\alpha}} = \frac{e^{-(\alpha-1)\varepsilon}}{\alpha-1}\Big(1-\frac{1}{\alpha}\Big)^{\alpha}.$$

The supremum is attained at $y = \alpha e^{\varepsilon}/(\alpha-1)$.

Conclusion. Hence $\mathbb{E}_Q[(P/Q - e^{\varepsilon})_+] \leq c_{\alpha}\mathbb{E}_Q[(P/Q)^{\alpha}] = c_{\alpha} e^{(\alpha-1)D_{\alpha}(P\|Q)} \leq c_{\alpha} e^{(\alpha-1)\alpha\rho}$. Take the infimum over α. Each term is continuous and increasing in ρ, so the infimum is nondecreasing and upper semicontinuous. It is also continuous here, because the infimum may be restricted to a compact set of α for ρ in a compact range. As $\rho \to 0$, taking $\alpha = 1 + \sqrt{\varepsilon/\rho}$ shows $\delta \to 0$ for $\varepsilon > 0$. □

Remark 5.5. The conversion is valid for every ρ-zCDP mechanism, but it is not the tightest possible for the Gaussian mechanism. For the example, $\varepsilon = 1$ and $\delta_{\mathrm{DP}} \approx 10^{-9}$ give $\rho_{\mathrm{run}} = 0.01496$. This agrees with the specification.

5.2 SIM-CDP

Let λ be the computational security parameter. $\mathrm{view}_{\mathsf{R}}(D)$ comprises three things: R's input (cohort identifiers and $\mathbf{X}_R$), its random tape, and every message it receives in one run on dataset D.

Definition 5.6 (SIM-CDP for R, after Mironov et al. [16])**.** The protocol is (ε,δ)-*SIM-CDP for* R *with statistical slack* γ if two objects exist:

- a randomized mechanism $\mathcal{M}$ that is (ε,δ)-DP with respect to $\sim$;
- a probabilistic polynomial-time simulator $\mathcal{S}$.

They must satisfy, for every dataset D and every PPT distinguisher $\mathcal{A}$,

$$\big|\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] - \Pr[\mathcal{A}(\mathcal{S}(\mathcal{M}(D))) = 1]\big| \leq \gamma + \mathrm{negl}(\lambda).$$

With $\gamma = 0$ this is the simulation-based notion of [16], which those authors state for pure ε [*citation detail unverified:* whether MPRV09 also define the (ε,δ) variant]. The extension to (ε,δ) is the standard one. The slack γ makes explicit the statistical terms that are not negligible in λ at concrete κ.

Lemma 5.7 (From SIM-CDP to indistinguishability)**.** *Suppose the protocol is (ε,δ)-SIM-CDP with slack γ. Then for all $D \sim D'$ and every PPT $\mathcal{A}$,*

$$\Pr[\mathcal{A}(\mathrm{view}(D)) = 1] \leq e^{\varepsilon}\Pr[\mathcal{A}(\mathrm{view}(D')) = 1] + \delta + (1+e^{\varepsilon})\gamma + \mathrm{negl}(\lambda).$$

That is, the protocol is $(\varepsilon, \delta + (1+e^{\varepsilon})\gamma)$-IND-CDP in the sense of [16].

Proof. Chain three steps: $\mathrm{view}(D) \approx \mathcal{S}(\mathcal{M}(D))$; then DP of $\mathcal{S} \circ \mathcal{M}$, which is post-processing; then $\mathcal{S}(\mathcal{M}(D')) \approx \mathrm{view}(D')$. □

Lemma 5.8 (Post-processing and composition of SIM-CDP)**.** *(a) If the protocol is (ε, δ)-SIM-CDP with slack γ, then so is $g(\mathrm{view}_{\mathsf{R}}(D))$ for every PPT g, with simulator $g \circ \mathcal{S}$.*

(b) Consider K runs on the same dataset, with independent keys and randomness and with R's inputs fixed. Suppose run k is simulated by $\mathcal{S}_k$ from a ρ_k-zCDP mechanism $\mathcal{M}_k$ with slack γ_k. Then the joint view is simulated by $(\mathcal{S}_k)_k$ from $(\mathcal{M}_k)_k$, which is $\sum_k \rho_k$-zCDP, with slack $\sum_k \gamma_k$.

In particular, the per-base-register ledger of the specification bounds the privacy loss of all runs, even if researchers collude. This holds when runs use fresh keys, which Algorithm 1 provides.

Proof. (a) is immediate. For (b), use a hybrid over runs: replace run k by $\mathcal{S}_k(\mathcal{M}_k(D))$ one run at a time. This is possible because run k uses keys and randomness independent of the other runs, so the reduction can simulate those runs itself. Then apply Lemma 5.2. If the researchers' inputs and the runs' parameters depend adaptively on earlier outputs, the hybrid is written out in Part C, the composition appendix (Theorem 4.2 there, `thm:main`): all smudging steps are done first in one statistical hybrid, the computational steps follow run by run, and a Rényi filter replaces Lemma 5.2. Part C also requires the ledger to charge a run's full cost at admission, since the δ_B part of an aborted run's unmade releases must stay charged, and gives the ledger with a hockey-stick slack (Corollary 5.4 there). □

5.3 Assumptions

Assumption 5.9 (RLWE for the re-randomizer)**.** Let $a \leftarrow \mathcal{R}_{q_0}$ be uniform, $u \leftarrow \chi_{\mathrm{key}}$ (uniform ternary) and $e_1 \leftarrow \chi_{\mathrm{err}} = \mathcal{N}_{\mathbb{Z}}(0, \sigma_e^2)^N$. Then $(a, au + e_1) \approx_c (a, \mathrm{U}(\mathcal{R}_{q_0}))$. This is decision-RLWE with one sample, a ternary secret and discrete Gaussian error with $\sigma_e = 3.2$. Let Adv_1 denote the best advantage.

Assumption 5.10 (RLWE with circular security for O's key shares)**.** Let $s_O \leftarrow \chi_{\mathrm{key}}$, let every error be drawn from χ_{err}, and let all a's be uniform common random strings. The tuple

$$\left(a,\ -as_O + e_O^{\mathrm{pk}};\ \{a_{g,j},\ -a_{g,j}s_O + P\mathsf{g}_j\psi_g(s_O) + e_{O,g,j}^{\mathrm{rk}}\}_{g\in\{g_0,\ldots,g_{L-1}\},\, j\le d_{\mathrm{num}}}\right)$$

is computationally indistinguishable from uniform over $\mathcal{R}_{Q_2} \times \mathcal{R}_{PQ_2}^{Ld_{\mathrm{num}}}$; let Adv_2 be the advantage. The first component is plain RLWE in s_O. The rotation-key components encrypt key-dependent messages $P\mathsf{g}_j\psi_g(s_O)$, so they need a circular (key-dependent-message) security assumption, as do all CKKS and BFV rotation keys.

These shares are exactly what R receives from O in the one-round public-key and rotation-key generation protocols of Mouchet et al. [17]. Under this assumption a semi-honest R's view of key generation is simulated by uniform elements; this is the "security of threshold key generation" used in the specification. [*citation detail unverified:* Mouchet et al. state that the rotation-key protocol relies on the same circular-security assumption as single-key RLWE schemes; we did not locate the exact statement number.]

The proof also uses three things proved above: Theorem 3.12 (Obligation O1), Lemma 2.3 with Corollary 2.4, and Proposition 5.3 with Proposition 5.4. It also uses O's honest behaviour in Algorithm 5:

- (u, e_0, e_1, η) are fresh and independent for each release;
- η is *exactly* $D_{\mathbb{Z}^N, \sigma_{c,s}}$;

- c'_0 is neither sent nor reused.

The proof does *not* use PSI security, the random-oracle model, or any assumption on the distribution of rounding errors.

5.4 The theorem

Theorem 5.11 (Output privacy against an honest-but-curious R; Theorem 1 of the specification, corrected)**.** *Assume (M1)–(M5), Assumptions 5.9 and 5.10, and that for every release s:*

- $\sigma_{c,s} \geq \max\{2^{\kappa} B_{\mathrm{tot},s}/\sqrt{2\pi},\, 2^{42}\}$, *with $B_{\mathrm{tot},s}$ from Theorem 3.12, or from Lemma 4.1 for OpenFHE's key switching (M3′) (the floor was 2^{20} in earlier versions; Remark 2.5);*
- $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$, *with $\widehat{\Delta}_{\mathrm{enc},s}$ from Corollary 3.13 (equal scales) or Lemma 4.3 (decoupled scales, as in the evaluation);*
- $\sum_s \rho_s \leq \rho_{\mathrm{run}}$, *with ρ_{run} from Proposition 5.4 at $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$.*

Then R's view of one run is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP for $\sim$, with statistical slack

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big)$$

and computational loss $n_{\mathrm{rel}}\mathrm{Adv}_1 + \mathrm{Adv}_2$. The mechanism is

$$\mathcal{M}(D) = \big(\omega_{\mathsf{R}},\ \Lambda,\ (f_s(D;\pi) + \eta_s)_s\big),$$

where ω_{R} is a fresh random tape for R, and Λ is R's view of the Link phase (Algorithm 2) generated with fresh coins for O. Those coins include the OPRF key, the dummy identifiers' labels and the permutation τ, hence π; the same τ fixes the arrangement used in $f_s(\cdot;\pi)$. By Lemma 5.7, the view is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}} + (1 + e^{\varepsilon_{\mathrm{run}}})\gamma)$-IND-CDP.

Proof. *$\mathcal{M}$ is ρ_{run}-zCDP, and hence $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP.* The pair $(\omega_{\mathsf{R}}, \Lambda)$ has the same law under D and D'. Neighbours differ only in $(\mathbf{x}_O, y)$ of one row, and neither the Link phase nor R's coins depend on $(\mathbf{x}_O, y)$: the labels are positions $\tau(j)$, and the identifiers, including the dummy ones, are part of the fixed part of the dataset. Conditionally on $(\omega_{\mathsf{R}}, \Lambda)$, the last component is the composition of the mechanisms $f_s(\cdot;\pi) + \eta_s$. Their sensitivity is at most $\widehat{\Delta}_{\mathrm{enc},s}$ for every arrangement π, by Corollary 3.13 or Lemma 4.3. Apply Proposition 5.3, Lemma 5.2 and Proposition 5.4.

The simulator. On input $(\omega_{\mathsf{R}}, \Lambda, (w_s)_s)$, $\mathcal{S}$ proceeds as follows:

1. output ω_{R} and Λ;
2. replace O's key-generation shares by uniform elements of the right rings;
3. for each release, sample $c^*_{1,s} \leftarrow \mathrm{U}(\mathcal{R}_{q_0})$ and output $(c^*_{1,s},\ w_s - c^*_{1,s}s_R \bmod q_0)$, reading s_R from ω_{R};
4. output the public parameters, including the outputs of CALIBRATE. These depend on public inputs only, because $B_{\mathrm{tot},s}$ is a worst-case bound.

Hybrids.

H0 The real view.

H1 Compute $h_{O,s}$ as $v_s - c'_{1,s}s_R$, with $v_s = c'_{0,s} + c'_{1,s}s + \eta_s \bmod q_0$. This is identical to H0, since $h_{O,s} = c'_{0,s} + c'_{1,s}s_O + \eta_s$.

H2$^{(s)}$ For $s = 1, \ldots, n_{\mathrm{rel}}$ in turn, replace v_s by $f_s(D;\pi) + \eta_s \bmod q_0$. By Proposition 3.8, $v_s \equiv f_s + e_{\mathrm{tot},s} + \eta_s$. Let Z be everything else in the hybrid. Then η_s is independent of $(Z, e_{\mathrm{tot},s})$, because it is fresh and used only in v_s. Corollary 2.4 with $B = B_{\mathrm{tot},s}$ and Theorem 3.12, which holds for every D and every value of the public randomness, bound the statistical distance between consecutive hybrids by $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$. Summing over s gives γ. Three precisions:

– *The side information includes* R*'s random tape.* Z contains ω_{R}, hence s_R, R's encryption errors and its key-share errors, all of which enter $e_{\mathrm{tot},s}$ (through the term of R's errors times O's data, the shares of e^{pk} and e^{rk}, and the s-dependent terms). Corollary 2.4 allows any joint law of $(Z, e_{\mathrm{tot},s})$, so the simulator needs only $\mathcal{M}(D)$, not the exact $f_s(D)$: this is simulation in the standard sense, not the relaxed sense of Hwang et al. [15] in which the simulator also receives the exact output.

– R*'s errors must be honestly sampled.* Theorem 3.12 bounds $e_{\mathrm{tot},s}$ under (M5), with every error drawn from χ_{err}. A researcher who inflated its encryption errors would push the term of R's errors times O's data beyond $B_{\mathrm{tot},s}$, and that term is correlated with O's data. This is part of the honest-but-curious model, and it must be stated as such.

– γ *is statistical.* The slack γ is a concrete statistical distance at the chosen κ and δ_B, not a quantity negligible in λ; it is accounted for explicitly in δ (Lemma 5.7).

H3$^{(s)}$ Replace $c'_{1,s}$ by a uniform element of $\mathcal{R}_{q_0}$. After H2, u_s and $e_{1,s}$ enter the view only through $c'_{1,s} = c_{1,s} + au_s + e_{1,s}$. The reason is that $c'_{0,s}$ is not sent, and neither v_s nor anything else uses $c'_{0,s}$, $e_{0,s}$, u_s or $e_{1,s}$; in particular the hint $e^{\mathrm{pk}}u_s + e_{1,s}s$ was removed with $e_{\mathrm{tot},s}$.

The reduction to Assumption 5.9 works as follows. On a challenge (a^*, b^*), it sets a to a uniform CRT lift of a^*, so that a is uniform at the full modulus. It samples s_O, all keys, all errors and all other releases itself; it knows D. It sets $c'_{1,s} = c_{1,s} + b^*$. Releases $s' < s$ are already uniform and releases $s' > s$ are real, and both are computable because the reduction knows a. The loss per release is Adv_1. Fresh u_s per release is essential here. If u were reused, then $c'_{1,s} - c'_{1,s'}$ would equal $c_{1,s} - c_{1,s'} + e_{1,s} - e_{1,s'}$. That is a short perturbation of a deterministic function of the data and the public randomness, and the hybrid would fail.

H4 Replace O's key-generation shares by uniform elements. After H3, the view depends on $(s_O, e_O^{\mathrm{pk}}, e_O^{\mathrm{rk}})$ only through these shares:

– $h_{O,s}$ no longer contains s_O;

– the ciphertexts M_s, and therefore the key-switching and conversion steps, no longer influence the view;

– $e_{\mathrm{tot},s}$ has been removed.

The reduction to Assumption 5.10 embeds its challenge as O's shares. It computes the joint public key and R's own shares from the challenge and ω_{R}, and runs the rest honestly. The loss is Adv_2.

Only $p_{0,O}$ is sent. In the protocol as implemented and as stated in the paper, O sends R only its public-key share $p_{0,O} = -as_O + e_O^{\mathrm{pk}}$; O sums the rotation-key shares itself and never sends its own. After H3 the rotation keys enter R's view only through the ciphertexts M_s, which no longer enter it. So H4 needs only the first component of Assumption 5.10: plain decision-RLWE with one sample in s_O, with no circular-security assumption. The rotation-key components are needed only if O's rotation-key shares are sent.

H4 is distributed exactly as $\mathcal{S}(\mathcal{M}(D))$, with the coins of $\mathcal{M}$ coupled to those of the hybrid: the Link part of H4 is generated exactly as Λ, and every other component matches the simulator. □

Remark 5.12 (Comparison with the specification's proof sketch)**.** The sketch's hybrids H0–H3 match ours, with these differences:

(i) the centre is $f_s(D)$, not $\mathsf{Ecd}(\mathbf{z}_s(D))$, which changes Δ_{enc} to $\widehat{\Delta}_{\mathrm{enc}}$;

(ii) the per-release distance is $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$;

(iii) the statistical slack is kept separate from δ, and it enters δ with the factor $1 + e^{\varepsilon}$ under the indistinguishability reading;

(iv) the sketch's H4 invokes PSI security, which is unnecessary under the specification's adjacency.

If one wants the simulator to use only the leakage G4, that is, $(I, \pi|_I)$, rather than a transcript

generated from the identifiers, then one needs the semi-honest receiver security of the labeled PSI [5]. That in turn needs circuit privacy of O's BFV replies (Correction 7).

Remark 5.13 (Why exactness of the sampler matters)**.** Proposition 5.3 needs η_s to be exactly $D_{\mathbb{Z}^N,\sigma}$, through Lemmas 2.1 and 2.2. A sampler at statistical distance ν from it adds ν per coordinate, so $N\nu$ per release, to γ. That is acceptable only if $N\nu \ll 2^{-\kappa}$. A floating-point sampler at $\sigma \approx 2^{52}$ is not of this kind. Float64 values of magnitude at least 2^{53} are spaced at least 2 apart, so the rounded output misses odd integers in that range. More generally, the low-order bits of its output are not smoothed. So ν is not small, and the Rényi divergence between neighbouring outputs can be infinite. This is the review's point, and it is why SampleDG must be exact (Section 8).

5.5 Release timing and the budgeted sampler

The view of Definition 5.6 contains the messages R receives, not the times at which they arrive. An honest-but-curious R can record those times, and the time O takes to answer a release request depends on how long the noise η_s took to draw. With the plain CKS20 sampler (Section 8) that time grows with $|\eta_s|$: its discrete Laplace and Bernoulli($e^{-\gamma}$) loops run a value-dependent number of times. This subsection replaces it by a sampler with a *fixed work budget*, proves that its timing depends on its output only on a failure event of probability at most ξ_T, and charges ξ_T per release to the statistical slack. Throughout, τ_s is the time from R's request for release s to the arrival of O's answer (the release or a refusal).

The sampler. Fix an integer variance $S = \sigma^2 \geq 1$, a number n of values, $r = 128$ bits per coin, $t = 2^b$ with b a public function of S (the implementation takes the b near $\frac{1}{2}\log_2 S$ that maximizes p_{lo} below), $C = \lceil 2^r S/(2t^2)\rceil$ and $c = C/2^r \geq S/(2t^2)$. Let $E_i = \lfloor 2^r e^{-i}\rfloor$ for $i \geq 1$ and $J_0 = \min\{i : E_i = 0\}$ ($J_0 = 89$). A *coin* is a uniform $U \in [0,1)$ of which only the first r bits $R = \lfloor 2^r U\rfloor$ are drawn. Public integers T, $J_v < J_0$ and $k_{\max}$ are fixed below.

Definition 5.14 (SampleDGBudget(S,n))**.** For $j = 1,\dots,T$, with fresh coins for each j:

(i) a sign bit s_j, u_j uniform on $\{0,\dots,t-1\}$, and $v_j = \#\{1 \leq i \leq J_v : R_j^v < E_i\}$ from a coin U_j^v;
(ii) $m_j = u_j + tv_j$, and $y_j = m_j$ if $s_j = 0$, $y_j = -(m_j+1)$ if $s_j = 1$;
(iii) $G_j = \lfloor 2^{r-1} y_j^2/S\rfloor - 2^r v_j + C$, computed exactly; $n_j = \lfloor G_j/2^r\rfloor$ and $F_j = G_j \bmod 2^r$;
(iv) $\alpha_j = 1$ iff $[n_j = 0$ or $R_j^a < E_{\min(n_j,J_0)}]$ and L_j is even, where L_j is the number of leading $k \in \{1,\dots,k_{\max}\}$ with $R_j^{(k)} < \lfloor F_j/k\rfloor$, from coins $U_j^a, U_j^{(1)},\dots,U_j^{(k_{\max})}$.

Trial j is *undecided* if $v_j = J_v$; or $R_j^v = E_i$ for some $i \leq J_v$; or $n_j \geq 1$ and $R_j^a = E_{\min(n_j,J_0)}$; or $R_j^{(k)} = \lfloor F_j/k\rfloor$ for some $k \leq k_{\max}$; or $L_j = k_{\max}$. The call *fails* (event F) if some trial is undecided or $A = \sum_{j\leq T}\alpha_j < n$; otherwise it returns the y_j of the first n trials with $\alpha_j = 1$.

All T trials are computed, whatever their outcome, and the surplus is discarded. Step (iv) is Bernoulli(e^{-n_j}) by one coin and Bernoulli(e^{-f_j}), $f_j = g_j - n_j$, by Algorithm 1 of CKS20 with exactly $k_{\max}$ coins, where

$$g_j = \frac{y_j^2}{2S} - v_j + c.$$

The ideal sampler. On the same probability space, extended by fresh coins, let the *ideal sampler* run an infinite sequence of trials with the same coins for $j \leq T$ and new ones for $j > T$, read every coin to full precision, and use no bound: $v_j^* = \#\{i \geq 1 : U_j^v < e^{-i}\}$, y_j^* from (s_j, u_j, v_j^*) as in (ii), g_j^* as above, and $\alpha_j^* = [U_j^a < e^{-\lfloor g_j^*\rfloor}] \cdot [K_j^* \text{ odd}]$, with K_j^* the first $k \geq 1$ such that $U_j^{(k)} \geq (g_j^* - \lfloor g_j^*\rfloor)/k$ (coins $U_j^{(k)}$ for $k > k_{\max}$ are fresh). Its output is the y_j^* of the first n trials with $\alpha_j^* = 1$.

Lemma 5.15 (The ideal sampler is exact, and its acceptances carry no information)**.** *The trials* $(y_j^*, \alpha_j^*)_{j\ge 1}$ *are i.i.d.,* $\Pr[\alpha_j^* = 1] = p = (1-e^{-1})e^{-c}\Theta/(2t)$ *with* $\Theta = \sum_{y\in\mathbb{Z}} e^{-y^2/(2S)}$*, and*

$$\Pr[y_j^* = y,\ \alpha_j^* = 1] = p \cdot \pi_S(y), \qquad \pi_S(y) = e^{-y^2/(2S)}/\Theta.$$

Hence the ideal output is exactly $\mathcal{N}_{\mathbb{Z}}(0,S)^n$ *and is independent of the acceptance sequence* $(\alpha_j^*)_{j\ge1}$*, which is i.i.d.* Bernoulli(p)*. Moreover* $p \ge p_{\mathrm{lo}} := (1-e^{-1})e^{-c}\sqrt{2\pi S}/(2t)$*.*

Proof. Proposal. v_j^* is geometric, $\Pr[v_j^* \ge k] = \Pr[U_j^v < e^{-k}] = e^{-k}$. The map $(s,u,v) \mapsto y$ of (ii) is a bijection from $\{0,1\} \times \{0,\dots,t-1\} \times \mathbb{N}$ onto $\mathbb{Z}$ (nonnegative y for $s=0$, negative for $s=1$), so $\Pr[y_j^* = y] = q(y) = (1-e^{-1})e^{-v(y)}/(2t)$ with $v(y) = \lfloor m(y)/t \rfloor$, $m(y) = y$ or $-y-1$.

Acceptance. $g(y) \ge 0$: in both cases $|y| \ge m(y) \ge tv(y)$, so $g(y) \ge t^2v^2/(2S) - v + S/(2t^2) = \frac{1}{2}(tv/\sigma - \sigma/t)^2 \ge 0$. Given y, the two factors of α_j^* use independent coins; the first is 1 with probability $e^{-\lfloor g \rfloor}$, and the second with probability $e^{-\{g\}}$, since $\Pr[K^* > k] = \prod_{i\le k}\{g\}/i = \{g\}^k/k!$ and $\Pr[K^* \text{ odd}] = \sum_{k\ge0}(-1)^k\{g\}^k/k!$ (CKS20, Algorithm 1). Hence $\Pr[\alpha_j^* = 1 \mid y_j^* = y] = e^{-g(y)}$ and

$$\Pr[y_j^* = y, \alpha_j^* = 1] = \frac{1-e^{-1}}{2t}\, e^{-v(y)} e^{-y^2/(2S)+v(y)-c} = \frac{(1-e^{-1})e^{-c}}{2t}\, e^{-y^2/(2S)}.$$

Summing over y gives p, and dividing gives π_S.

Independence. The trials use disjoint coins, so they are i.i.d. Since the joint law of (y_j^*, α_j^*) factorizes as above, conditionally on the whole sequence $(\alpha_j^*)_j$ the values y_j^* of the accepted trials are i.i.d. π_S; so the output, the first n of them, is $\mathcal{N}_{\mathbb{Z}}(0,S)^n$ given every acceptance sequence.

Lower bound. By Poisson summation, $\Theta = \sqrt{2\pi S}\sum_{k\in\mathbb{Z}} e^{-2\pi^2 S k^2} \ge \sqrt{2\pi S}$. □

Lemma 5.16 (Coupling)**.** *Outside* F*, the budgeted sampler returns exactly the ideal sampler's output, and* $A = A^* := \sum_{j\le T}\alpha_j^*$*.*

Proof. Let trial j be decided. For a threshold $E = \lfloor 2^r x \rfloor$ and a coin with $R \ne E$: if $R < E$ then $U < (R+1)/2^r \le E/2^r \le x$; if $R > E$ then $U \ge R/2^r \ge (E+1)/2^r > x$. So $[R < E] = [U < x]$. Applied with $x = e^{-i}$, $i \le J_v$: the indicators defining v_j and v_j^* agree for $i \le J_v$, and $v_j < J_v$ gives $U_j^v \ge e^{-J_v}$, so they agree for all i, and $v_j = v_j^*$, $y_j = y_j^*$. $G_j = \lfloor 2^r g_j \rfloor$ exactly, because $2^r v_j$ and C are integers; hence $n_j = \lfloor g_j \rfloor$, $F_j = \lfloor 2^r\{g_j\} \rfloor$ and $\lfloor F_j/k \rfloor = \lfloor 2^r\{g_j\}/k \rfloor$. Applied with $x = e^{-n_j}$ (and, for $n_j \ge J_0$, with $E = 0$, where $R \ne 0$ gives $U \ge 2^{-r} > e^{-n_j}$) and with $x = \{g_j\}/k$ for $k \le k_{\max}$, the decisions agree with the ideal ones; $L_j < k_{\max}$ means that $K_j^* = L_j + 1 \le k_{\max}$, so the parity is decided by the coins drawn. Hence $\alpha_j = \alpha_j^*$ for every $j \le T$, $A = A^*$, and $A \ge n$ means that the ideal sampler's first n acceptances lie among the first T trials. □

Lemma 5.17 (Failure bound)**.**

$$\Pr[\mathsf{F}] \le \xi_{\mathrm{bound}} := \exp\Big(-T\,\mathrm{KL}\Big(\frac{n-1}{T}\ \Big\|\ p_{\mathrm{lo}}\Big)\Big) + T\Big(e^{-J_v} + \frac{1}{k_{\max}!} + (J_v + 1 + k_{\max})2^{-r}\Big) \quad \textit{whenever } n-1 < Tp_{\mathrm{lo}},$$

with $\mathrm{KL}(a\|p) = a\ln\frac{a}{p} + (1-a)\ln\frac{1-a}{1-p}$. Calibrate *(implementation:* `sampler.budget`*) takes* T *the smallest integer whose first term is at most* $\xi_T/2$*,* $J_v = \lceil \ln(8T/\xi_T) \rceil$ *and* $k_{\max}$ *the smallest with* $T/k_{\max}! \le \xi_T/8$*, and checks that the tie term is at most* $\xi_T/8$*; then* $\xi_{\mathrm{bound}} \le \xi_T$*.*

Proof. If no trial is undecided, $A = A^*$ by Lemma 5.16, so $\mathsf{F} \subseteq \{A^* < n\} \cup \bigcup_j \{j \text{ undecided}\}$. $A^* \sim \mathrm{Binomial}(T,p)$ with $p \ge p_{\mathrm{lo}}$ (Lemma 5.15); the lower tail is nonincreasing in p, and the Chernoff bound $\Pr[\mathrm{Bin}(T,p) \le k] \le e^{-T\mathrm{KL}(k/T\|p)}$ for $k < Tp$ gives the first term. For one trial: $v_j = J_v$ requires $R_j^v < E_{J_v}$, of probability $E_{J_v}2^{-r} \le e^{-J_v}$; $L_j = k_{\max}$ requires $U_j^{(k)} < \{g_j\}/k$ for all

$k \le k_{\max}$, of probability at most $1/k_{\max}!$; and each of the $J_v + 1 + k_{\max}$ ties is a coin equal to a threshold computed from other coins, of probability 2^{-r}. A union bound over the T trials completes the proof. The constants are certified: E_i is computed from rational bounds on e^{-1} (consecutive partial sums of its alternating series) with directed rounding at 768 bits, and both bounds must give the same floor. □

At $\xi_T = 2^{-80}$, $N = 2^{14}$ and the largest σ_c^2 of the 10^6-row runs ($\sigma_c^2 = 2^{136.7}$, $\sigma_c \approx 2^{68.4}$, quoted as $2^{68.5}$ elsewhere): $b = 68$, $p_{\text{lo}} = 0.4457$, $T = 39{,}088$ $(2.39N)$, $J_v = 69$, $k_{\max} = 29$, $\xi_{\text{bound}} = 2^{-80.9}$. For the flooding of SanitizeBFV ($\sigma_\psi = 2^{61}$, $N_B = 2^{14}$): $p_{\text{lo}} = 0.4805$, $T = 36{,}189$, the same J_v and $k_{\max}$, $\xi_{\text{bound}} = 2^{-80.9}$.

The cost assumption. A side channel needs a model of what the observer measures. We use the following.

Assumption 5.18 ((W): work, not values, determines time)**.** The time O takes to execute a sequence of operations is $\Phi(\varpi, Z)$, where the *trace* ϖ lists the operations together with their public shapes (array lengths, numbers of limbs, the native function called and the sizes of its operands, but not the operand values), Z is machine noise (scheduling, other processes, the state of caches and of the allocator) independent of every secret of the run and of D, and Φ does not depend on the operand values.

Lemma 5.19 (The trace of a call)**.** *In the implementation (`sampler.sample_residues`), the trace of a call that does not fail is a function of (S, n, ξ_T), the moduli and A only.*

Proof (by inspection of the code). The call draws $4T(1 + L_y + 8 + 4k_{\max})$ bytes from the CSPRNG (L_y limbs of 32 bits hold $|y| < 2^{b+\lceil \log_2(J_v+2)\rceil+1}$), and then executes a fixed sequence of numpy operations on arrays of shape $(\cdot, T)$ with limb counts that are functions of $(S, T, J_v, k_{\max})$: the comparisons for v against the J_v table values, the placement of v at bit b, the limb products y^2, $y^2\lfloor 2^P/S\rfloor$ and $\hat{Q} \cdot S$, one correction, the addition of C and subtraction of v at limb 4, a gather from the $(J_0 + 1)$-entry table, and $k_{\max}$ long divisions of F by k (by multiplication with $\lceil 2^{28}/k\rceil$ on 16-bit digits, not by a hardware division) and comparisons. No Python branch depends on a value. The only shape that varies is that of the index array of accepted trials (length A), from which the first n are taken; the residues of these n values modulo each q are computed by a fixed reciprocal multiplication and one correction. The failure test is evaluated in every call. □

Lemma 5.20 (Release timing)**.** *Under (W), for each release s there is a random variable τ_s^*, whose law depends only on the public parameters and the machine, which is independent of R's view with exact noise (and of D), and such that, with V the rest of R's view (its tape and messages),*

$$\mathrm{SD}\big((V, \tau_s),\ (V^*, \tau_s^*)\big) \le \xi_T,$$

where V^* is V with η_s replaced by the ideal sampler's output on the same coins, which is exactly $D_{\mathbb{Z}^N, \sigma_{c,s}}$ and independent of everything else in the run.

Proof. O's work for release s after the request is: the sampler call, the re-randomized partial decryption (a native call on fixed-shape polynomials), the deadline test and the hold, and the send. Let $\varpi(A)$ be the trace of a non-failing release with A accepted trials (Lemma 5.19; the other steps have fixed traces). Set $\tau_s^* = \Psi(\Phi(\varpi(A^*), Z), Z')$, where Ψ applies the deadline rule (hold until D, or refuse if the busy time exceeds D) and Z' is the network and scheduling noise of the send. On F^c, η_s equals the ideal output and $A = A^*$ (Lemma 5.16), so $(V, \tau_s) = (V^*, \tau_s^*)$; hence the statistical

distance is at most $\Pr[\mathsf{F}] \le \xi_T$ (Lemma 5.17). A^* is a function of the acceptance sequence, which is independent of the ideal output (Lemma 5.15) and of every other coin of the run; Z, Z' are independent of all secrets by (W); and D, the deadline, is fixed before the first release from timed dummy calls on public parameters with independent coins. So τ_s^* is independent of V^* and of D. □

Corollary 5.21 (Output privacy with release timing; the charge)**.** *Under the hypotheses of Theorem 5.11 and (W), with every release drawn by* SampleDGBudget *at failure bound ξ_T,* R*'s view of one run* including the arrival times $(\tau_s)_s$ *is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP with statistical slack*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97} + \xi_T\big),$$

for the mechanism $\mathcal{M}(D)$ of Theorem 5.11 extended by the independent, data-independent component $(\tau_s^*)_s$. In the hockey-stick ledger of Part C, the run's additive slack becomes $n_{\mathrm{rel}}(\delta_B + \delta' + \xi_T)$.

Proof. Add a hybrid H0′ before H1: replace, for $s = 1, \dots, n_{\mathrm{rel}}$ in turn, (η_s, τ_s) by (η_s^*, τ_s^*). Each step costs at most ξ_T by Lemma 5.20 (the lemma holds conditionally on everything outside release s, because the sampler's coins are fresh). After H0′ every η_s is exact and independent of the rest, which is what H2 uses, and the τ_s^* are independent of everything else and have a law that does not depend on D. The remaining hybrids are unchanged and carry $(\tau_s^*)_s$ along. The extended mechanism is still ρ_{run}-zCDP: a component independent of D and of the rest does not change any Rényi divergence (Lemma 5.2), and the simulator outputs it from $\mathcal{M}(D)$. The $n_{\mathrm{rel}}\xi_T$ is a statistical distance introduced before any other hybrid, exactly like δ_B; in Part C it enters the purely additive slack sum Γ (the proofs of Theorem 5.2 and Corollary 5.4 there treat every such term identically), and by Lemma 5.7 it is converted with the factor $1 + e^{\varepsilon}$. □

Remark 5.22 (The deadline is defence in depth; the overrun is not charged)**.** The theorem rests on the work budget, not on the padding. Under (W) the busy time of a release is a function of (A^*, Z), so whether it overruns the deadline is independent of η_s and of D: an overrun refusal is part of τ_s^* and needs no charge, whatever the machine load. The padding still helps where (W) is only approximately true: residual value-dependent costs are hidden below D, and only an overrun could expose them.

Remark 5.23 (The flooding of SanitizeBFV)**.** The flooding vectors of the PSI replies are drawn by the same sampler, one call per vector of N_B values ($2\alpha F$ calls per cuckoo table, n_{calls} in total). By Lemmas 5.16–5.20 applied to each call, the reply timing makes the view of the Link phase differ by at most $n_{\mathrm{calls}}\xi_T$ from one in which every flooding vector is exact and independent of the timing, which is what the smudging step of G4's second layer (Section 7) needs; a failed call aborts the run, inside the same bound. This adds $n_{\mathrm{calls}}\xi_T$ to the statistical distance of the second layer, not to δ: neighbouring datasets have the same identifiers and positions, so the Link phase, timing included, has the same law under both (Theorem 5.11). Setup checks $n_{\mathrm{calls}}\xi_T \le \delta_{\mathrm{psi}}$; at 10^6 rows ($\alpha = 5$, $F = 2$, one table) it is $20 \cdot 2^{-80} = 2^{-75.7}$.

Remark 5.24 (What (W) asks of the implementation, and what is not verified)**.** (W) is an assumption about the platform, not a theorem. The implementation is written so that only operations whose cost plausibly does not depend on values remain: fixed-shape numpy loops over `uint64` ($+, -, \times$, shifts, bitwise operations, comparisons, boolean reductions), with no hardware division (whose latency depends on its operands on many x86 cores) and no Python branch on a value; ChaCha20 from OpenSSL for the coins; and OpenFHE's NTT-based polynomial arithmetic. Not verified: that numpy's loops have no value-dependent fast paths on this machine; the gather from the 2.9 kB table (E_i), whose addresses depend on n_j (a cache channel, invisible over a network

but not on O's machine); the CPython allocator; and OpenFHE's own samplers for (u, e_0, e_1), which (CT) of Section 8 also covers. The measurements in `reports/v31_timing/` support (W) for the sampler: on coins forced to the extreme values ($v = 0$ everywhere, $v = J_v - 1$ everywhere, $u = 0$ everywhere) one call's trials take the same time as on random coins to within the machine's noise (medians within 6% in two runs of 30 and 100 calls per kind, at a load average near 20 from other jobs, with an ordering of the kinds that did not repeat), while the plain CKS20 sampler's median time per value grows with $|\eta|/\sigma$, from 9.7 μs below 1σ to 25.6 μs beyond 3σ. This is evidence, not a proof.

6 Input privacy (G1) and cohort privacy (G2)

O's view. Besides its own input D and random tape, O receives the following from R:

(i) *Key-generation shares:* $p_{0,R} = -as_R + e_R^{\mathrm{pk}}$, and the rotation-key shares $-a_{g,j}s_R + P\mathsf{g}_j\psi_g(s_R) + e_{R,g,j}^{\mathrm{rk}}$.

(ii) *Link messages:* the blinded OPRF queries, exactly $n_R^{\max}$ of them; R's BFV public key and any evaluation keys used by the PSI; and the BFV ciphertexts of windowed powers. Their number is fixed by public PSI parameters and $n_R^{\max}$.

(iii) *Uploads:* $(c_{0,h}, \varsigma_h)$ for each of the $(d_R + 2)m$ chunks, with $c_{0,h} = -a_h s_R + e_h + m_{U,h}$ and $a_h = H(\varsigma_h)$.

(iv) *Control messages:* the requests for the two or three releases. Their schedule is fixed by Algorithm 6 and does not depend on R's data. R's refusal at line 7 happens after the last release and is not sent to O.

O receives no decryption share and no ciphertext component that involves s_R other than these.

Assumption 6.1 (For G1 and G2)**.** (a) *RLWE with circular security in s_R.* Let $s_R \leftarrow \chi_{\mathrm{key}}$, let every error be drawn from χ_{err}, and let all a's be uniform. Then the tuple consisting of

- R's public-key share,
- its Ld_{num} rotation-key shares, and
- $(d_R + 2)m$ samples $(a_h, -a_h s_R + e_h)$ at modulus Q_2

is pseudorandom. This is multi-sample RLWE plus the circular assumption for the rotation-key shares. Samples at modulus Q_2 are reductions of samples at modulus PQ_2, so a single modulus suffices.

(b) *Random oracle.* The seed expansion H is a random oracle, and seeds are λ-bit strings drawn uniformly.

(c) *The PSI's encryption.* The PSI's BFV is IND-CPA under RLWE, with circular security for any relinearization key R sends.

(d) *The OPRF.* The blinded OPRF queries of [5] are distributed independently of the queried identifiers. For the Diffie–Hellman-type OPRF this holds perfectly in a prime-order group. [*citation detail unverified:* the concrete OPRF instantiation in [5] and whether its query blinding is perfectly or computationally hiding]

Theorem 6.2 (G1: R-input privacy)**.** *Under Assumption 6.1 there is a PPT simulator $\mathcal{S}_O$ with the following property. Given O's input and random tape, $(n_R^{\max}, \bar{n}_O)$ and the public parameters, $\mathcal{S}_O$ outputs a view that is indistinguishable from O's real view, for every input of R. The distance is at most*

$$\mathrm{Adv}_{\mathrm{RLWE},R} + \mathrm{Adv}_{\mathrm{BFV}} + q_H^2/2^\lambda + p_{\mathrm{cuckoo}},$$

where q_H is the number of oracle queries and p_{cuckoo} is the probability that R's cuckoo hashing fails in a way that O can observe, for example as an abort or a retry (bounded in Lemma 7.5). R's abort

on a detected false match is observable too; its probability, at most p_{false} (the paper's Remark on collisions), is added to the slack in the same way.

Proof. The simulator. $\mathcal{S}_O$ outputs:

- uniform key-generation shares;
- $n_R^{\max}$ uniform group elements as the OPRF queries;
- a fresh BFV public key, any evaluation keys, and encryptions of zero in the public number;
- $(d_R + 2)m$ pairs $(c^*_{0,h}, \varsigma^*_h)$, with $c^*_{0,h}$ uniform in $\mathcal{R}_{Q_2}$ and ς^*_h uniform seeds;
- the fixed control messages.

Hybrids.

1. Replace the OPRF queries by uniform group elements. The distance is zero for a perfectly hiding blinding, and the OPRF advantage otherwise.
2. Replace R's BFV ciphertexts of powers by encryptions of zero, at cost Adv_{BFV}.
3. In the random-oracle model, the $a_h = H(\varsigma_h)$ are uniform and independent unless two seeds collide, which happens with probability at most $q_H^2/2^\lambda$. The reduction programs $H(\varsigma_h)$ to the a-components of its RLWE challenge. The tuple of R's key shares and the values $-a_h s_R + e_h$ is then replaced by uniform elements, at cost $\text{Adv}_{\text{RLWE},R}$ by Assumption 6.1(a). Adding the known plaintext $m_{U,h}$ to a uniform element leaves it uniform.
4. The real protocol leaks the cuckoo-hashing outcome. Removing that leak costs at most p_{cuckoo}.

The final hybrid is $\mathcal{S}_O$'s output. O's own messages, such as the OPRF evaluations, PSI replies and releases, are functions of its view, so they need no simulation. □

Corollary 6.3 (G2: cohort privacy)**.** *$\mathcal{S}_O$ does not use R's cohort, n_R or $\mathbf{X}_R$. Hence, for any two cohorts of size at most $n_R^{\max}$, O's views are indistinguishable, up to twice the distance of Theorem 6.2. Padding the OPRF queries to exactly $n_R^{\max}$ is what makes the number of OPRF calls cohort-independent. The per-run fresh OPRF key and the quota are not needed against an honest-but-curious O; they matter only for linkage across runs.*

Remark 6.4. (i) The statement "O never receives a decryption share from R" is what makes G1 a pure encryption argument. If any future version sent R's share $c_1 s_R + e$, G1 would need its own flooding.

(ii) The random-oracle model is used only to treat the expanded a_h as uniform. A standard-model alternative is to send a_h in full, at twice the upload size.

(iii) G1 and G2 hold against an honest-but-curious O only. A deviating O could, for example, choose its PSI replies so as to learn the outcome of R's decryption through R's later behaviour. This is outside the model.

7 Disclosure (G4), the cuckoo-hashing slack and the reply noise

This section proves the first layer of G4 (what R learns from the Link phase) in the random-oracle model under CDH, bounds the cuckoo-hashing failure probability that enters the slack of G1 and G2, and bounds the BFV evaluation noise of the PSI replies in an explicit model of BFV. The first result was first written in the companion book (Chapter 8, Lemma `aln.keyed` and Theorem `aln.R`) and was checked here; the other two are new. They replace Correction 7 as far as the first layer is concerned.

The Link phase. O draws a fresh OPRF key $k \leftarrow \mathbb{Z}^*_{q_{\mathbb{G}}}$ and a uniform permutation τ of its $\bar{n}_O$ padded rows. Row j gets $y_j = F_k(\text{id}_j) = H_2(\text{id}_j, H_1(\text{id}_j)^k)$; its item, bins and label key are

domain-separated hashes of the two halves of y_j, and its label is $\tau(j)$ padded with the label key modulo t. R pads its identifiers to a list $\bar{X}$ of exactly $n_R^{\max}$, obtains F_k on $\bar{X}$ through the blinded protocol ($\alpha = H_1(x)^r$, $\beta = \alpha^k$), cuckoo-hashes its items, and sends BFV encryptions of the windowed powers under its own key. O evaluates the masked membership and label polynomials of each partition and replies with SanitizeBFV of the results. R's output is $(I, \pi|_I)$, or an abort.

Assumption 7.1 (CDH in the OPRF group)**.** Given (g, g^k, h) with k and h uniform, no PPT algorithm (with non-uniform advice) outputs h^k with probability above $\mathrm{Adv}_{\mathrm{CDH}}$.

Lemma 7.2 (OPRF-keyed items and labels are pseudorandom outside the match)**.** *Model H_1, H_2 and the derived hash functions as random oracles, let* R *be semi-honest, and let the distinguisher make at most q_{H_2} queries to H_2. Let J be the set of* O*'s rows whose identifiers are not in $\bar{X}$. Then* R*'s view of the OPRF phase together with $(F_k(\mathrm{id}_j))_{j\in J}$ changes by at most $q_{H_2}\mathrm{Adv}_{\mathrm{CDH}} + \epsilon_{\mathrm{coll}}$ in distinguishing advantage when these outputs are replaced by independent uniform strings, for every input; $\epsilon_{\mathrm{coll}} \le \bar{n}_O n_R^{\max} 2^{-\ell_{\mathrm{id}}} + (q_{H_1} + \bar{n}_O)/q_{\mathbb{G}}$ accounts for a collision of a fresh dummy identifier of length ℓ_{id} with the other list and for H_1 hitting the identity.*

Proof. Identical until a bad event. In the real experiment $F_k(\mathrm{id}_j) = H_2(\mathrm{id}_j, H_1(\mathrm{id}_j)^k)$; in the modified one it is a fresh uniform string, and H_2 is answered uniformly elsewhere. An honest R queries H_2 only at points $(x, H_1(x)^k)$ with $x \in \bar{X}$, so the two experiments are identical unless the distinguisher queries H_2 at $(\mathrm{id}_j, H_1(\mathrm{id}_j)^k)$ for some $j \in J$ (event Bad), which has the same probability in both.

A CDH solver. Given (g, g^k, h), run the modified experiment with O's input fixed (non-uniform advice), programming $H_1(x) = g^{a_x}$ at every queried point other than O's identifiers in J, and $H_1(\mathrm{id}_j) = h\, g^{b_j}$ for $j \in J$, with fresh uniform exponents. Answer a blinded query $\alpha = g^{a_x r}$ with $(g^k)^{a_x r} = \alpha^k$; compute the outputs of the matched rows as $H_2(x, (g^k)^{a_x})$ and those of the rows in J as fresh strings. Preprocessing, evaluation and sanitization use no other secret, and R's own coins are sampled by the solver. At the end, pick a uniformly random H_2 query (id_j, Z) with $j \in J$ and output $Z\,(g^k)^{-b_j}$. On Bad some query has $Z = h^k (g^k)^{b_j}$, picked with probability at least $1/q_{H_2}$. So $\Pr[\mathsf{Bad}] \le q_{H_2}\mathrm{Adv}_{\mathrm{CDH}}$; the programmed values are uniform except on the events counted in ϵ_{coll}. □

Theorem 7.3 (G4, first layer: R learns only the match and its positions)**.** *Under the hypotheses of Lemma 7.2 there is a PPT simulator $\mathcal{S}_\Lambda$ that, given* R*'s input, its output $(I, \pi|_I)$ and the public sizes, outputs* R*'s view of the Link phase up to distinguishing advantage $q_{H_2}\mathrm{Adv}_{\mathrm{CDH}} + \epsilon_{\mathrm{coll}}$, for every input. Jointly with the output, the distance grows by the probability of the correctness failures on which* R*'s output differs from $(I, \pi|_I)$ (overflow of a bin, a cuckoo failure, a false match; Lemma 7.5 and the paper's Remark on collisions). The statement needs neither circuit privacy of the BFV evaluation nor* SanitizeBFV*, nor a bound on the reply noise.*

Proof. $\mathcal{S}_\Lambda$ samples R's tape, runs R's code to produce the padded list, the blinded elements, the BFV keys and the encrypted windowed powers; draws a key k^* and answers the blinded elements with it; creates one row per $i \in I$ with output $F_{k^*}(i)$ (computed by R's code) and position $\pi(i)$, and $\bar{n}_O - |I|$ rows with uniform outputs and positions forming a uniform bijection onto $[\bar{n}_O] \setminus \pi(I)$; and runs O's preprocessing (partition placement included), evaluation, masking and sanitization with fresh coins. By Lemma 7.2 the real view is within the stated distance of the experiment in which O's rows outside the match carry uniform outputs. In that experiment the key enters only through R's own OPRF answers, so k^* may replace k; positions outside the match are a uniform bijection onto the complement of $\pi(I)$ because τ is uniform; the labels outside the match are padded by uniform keys. Every reply, its noise included, is a fixed randomized function of R's ciphertexts and

of these inputs, so the two experiments coincide. Overflow aborts, cuckoo failures, false matches and the partition index of a match arise from identically distributed items and bins and are reproduced exactly. □

Remark 7.4 (Second reading; what remains of Correction 7)**.** The book's lemma omits the collision term ϵ_{coll}, which is negligible; otherwise its proof is correct as written. The reduction is non-uniform (it embeds O's input), so Adv_{CDH} is against non-uniform adversaries. Circuit privacy is not needed for the first layer: whatever the noise of a reply reveals is a function of pseudorandom items and labels. The second layer, SanitizeBFV, makes the reply noise independent of O's data up to a statistical distance, given a worst-case bound B_ψ (Proposition 7.7 below bounds its evaluation part in a model of BFV).

Lemma 7.5 (The cuckoo-hashing slack)**.** *Let n items each have h candidate bins drawn independently and uniformly from m bins (in the random-oracle model for the bin hashes; with $m = 2^{14}$ dividing 2^{256} the reduction modulo m of a uniform hash is exactly uniform). The probability that no assignment of the items to distinct candidate bins exists is at most*

$$p_{\text{noassign}}(n, m, h) \le \sum_{s=2}^{n} \binom{n}{s} \binom{m}{s-1} \left(\frac{s-1}{m}\right)^{hs}.$$

An insertion procedure that finds an assignment whenever one exists (augmenting paths, or the implementation's random walk followed, on failure, by an augmenting-path search) fails with exactly this probability, so $p_{\text{cuckoo}} \le p_{\text{noassign}}$. At $m = 2^{14}$ and $h = 3$ the bound is $1.7 \cdot 10^{-15} = 2^{-49.1}$ for $n = 2{,}000$ (the evaluation's $n_R^{\max}$, one table), $2^{-47.0}$ for $n = 4{,}096$, $2^{-45.0}$ for $n = 8{,}192$, and at most 2^{-40} for every $n \le 10{,}472$ (load 0.639).

Proof. By Hall's theorem an assignment exists unless some set S of $s \ge 2$ items has fewer than s candidate bins in total (a single item always has a bin). Fix S and a set T of $s-1$ bins; all hs candidates of S lie in T with probability $((s-1)/m)^{hs}$. A union bound over S and T gives the sum (a set with fewer than $s-1$ bins is contained in some T). The numbers are from `experiments/proofs_check/tv1_cuckoo.py`, which sums every term. □

Remark 7.6 (What is not covered)**.** The implementation's insertion is a random walk with at most 2,000 evictions per item. Until 2026-10-25 it had no fallback, and its failure probability when an assignment exists was not bounded. It now completes the walk's partial assignment by augmenting paths when the walk exceeds its step limit (`implementation/vfr/psi.py`, `cuckoo` and `_augment`), so it finds an assignment whenever one exists and $p_{\text{cuckoo}} = p_{\text{noassign}}$. The union bound is useless above load about 0.64 at $m = 2^{14}$ (the implementation allows tables up to load 0.9, used only when $n_R^{\max} > 10{,}472$).

7.1 The BFV evaluation noise of the replies

We bound the evaluation part $w_{\text{eval}} = (q'/q_B)v$ of the switched reply error in the following model of BFV (with R's ternary key s_B, plaintext modulus t, $\Omega_{\text{B}} = \lfloor q_B/t \rfloor$, $r_t = q_B - t\Omega_{\text{B}} \in [0, t)$, ring dimension $N = N_B$, and every integer polynomial represented with centred coefficients).

(B1) A ciphertext (c_0, c_1) with plaintext m, $\|m\|_\infty \le t/2$, satisfies $c_0 + c_1 s_B = \Omega_{\text{B}} m + v + q_B k$ as integer polynomials; $\|c_i\|_\infty \le q_B/2$ and $\|v\|_\infty \le q_B/2$ give $\|k\|_\infty \le \kappa_k := (N+3)/2$.

(B2) Every error coefficient has absolute value at most B_e (OpenFHE's table sampler at $\sigma = 3.19$ saturates at $B_e = 27$), so R's fresh encryptions have $\|v\|_\infty \le E_0 := (2N+1)B_e$: $v = e_B^{\text{pk}} u' + e_0 + e_1 s_B$ for a public-key encryption, and $v = e$, with $\|v\|_\infty \le B_e$, for the secret-key encryption that the implementation uses.

(B3) A ciphertext product computes $\tilde{d}_i = \lfloor t\, d_i/q_B \rceil$ from the tensor $(d_0, d_1, d_2) = (c_0c_0',\ c_0c_1' + c_1c_0',\ c_1c_1')$, with rounding errors of sup-norm at most $\frac{1}{2}$ up to an additive error of sup-norm at most E_{rns} (an allowance for an RNS implementation), and relinearizes $\tilde{d}_2$, adding an error of sup-norm at most E_{rl}.

(B4) A plaintext multiplication multiplies both components by a plaintext polynomial a with $\|a\|_\infty \le t/2$; a plaintext addition adds $\Omega_{\mathrm{B}} a$ to c_0.

Proposition 7.7 (The reply's evaluation noise in the BFV model)**.** *Under (B1)–(B4), let* $F(E,E') = \frac{Nt^2}{4} + t(\frac{Nt}{4} + \frac{1}{2}) + \frac{Nt}{2}(E+E') + \frac{tN}{q_B}EE' + Nt^2\kappa_k + tN\kappa_k(E+E') + \frac{1+N+N^2}{2} + E_{\mathrm{rl}} + E_{\mathrm{rns}}$*. A product of two ciphertexts with noise sup-norms at most* E *and* E' *(both below* $q_B/(2t)$*) has noise sup-norm at most* $F(E,E')$*. Hence, if every power* $\mathbf{w}^d$*,* $d \le D$*, is a product tree of depth at most* $L_\times$ *over the fresh windows, its noise is at most* $E_{L_\times}$ *with* $E_{l+1} = F(E_l, E_l)$*, and a reply* $\sum_{d=0}^{D} a_d \odot \mathrm{ct}(\mathbf{w}^d)$ *(with* $\mathrm{ct}(\mathbf{w}^0)$ *the plaintext* 1*) has noise at most*

$$E_{\mathrm{eval}} = D\Big(\tfrac{Nt}{2}E_{L_\times} + t\big(\tfrac{Nt}{4} + \tfrac{1}{2}\big)\Big) + \tfrac{t(D+1)}{2},$$

so that $\|w_{\mathrm{eval}}\|_2 \le \sqrt{N}\,(q'/q_B)\,E_{\mathrm{eval}}$*, provided* $E_{\mathrm{eval}} < q_B/(2t)$ *(which also gives correct decryption before the switch).*

Proof. Product. Multiply the two invariants of (B1) and scale by t/q_B. With $t\Omega_{\mathrm{B}} = q_B - r_t$ and $mm' = [mm']_t + t\mu$, $\|\mu\|_\infty \le Nt/4 + \frac{1}{2}$, the identity

$$\begin{aligned}\tfrac{t}{q_B}(\Omega_{\mathrm{B}}m + v + q_Bk)(\Omega_{\mathrm{B}}m' + v' + q_Bk') \equiv \Omega_{\mathrm{B}}[mm']_t - \tfrac{r_t}{q_B}\Omega_{\mathrm{B}}mm' - r_t\mu \\ + \tfrac{t\Omega_{\mathrm{B}}}{q_B}(mv' + m'v) + \tfrac{t}{q_B}vv' - r_t(mk' + m'k) + t(vk' + v'k)\end{aligned}$$

holds modulo q_B (the terms $t\Omega_{\mathrm{B}}(mk' + m'k)$ and tq_Bkk' reduce to $-r_t(mk' + m'k)$ and 0). The rounded tensor adds $r_0 + r_1s_B + r_2s_B^2$, of sup-norm at most $\frac{1}{2}(1 + \|s_B\|_1 + \|s_B^2\|_1) \le \frac{1}{2}(1 + N + N^2)$, plus E_{rns}; relinearization adds E_{rl}. Bound each term with $\|ab\|_\infty \le N\|a\|_\infty\|b\|_\infty$: $\frac{r_t\Omega_{\mathrm{B}}}{q_B} \le 1$ and $\|mm'\|_\infty \le Nt^2/4$; $r_t < t$; $\|mv' + m'v\|_\infty \le \frac{Nt}{2}(E+E')$; $\|vv'\|_\infty \le NEE'$; $\|mk' + m'k\|_\infty \le Nt\kappa_k$; $\|vk' + v'k\|_\infty \le N\kappa_k(E+E')$. The sum is $F(E,E')$. The bound is increasing in E and E', so a tree of depth l has noise at most E_l.

Evaluation. $a \odot (c_0, c_1)$ has invariant $a(\Omega_{\mathrm{B}}m + v + q_Bk) \equiv \Omega_{\mathrm{B}}[am]_t - r_t\mu_a + av$, with $\|\mu_a\|_\infty \le Nt/4 + \frac{1}{2}$ and $\|av\|_\infty \le \frac{Nt}{2}\|v\|_\infty$. Summing D such terms and the plaintext term a_0, the messages add up to at most $(D+1)t/2$ in sup-norm; reducing the sum modulo t changes the noise by $r_t\nu$ with $\|\nu\|_\infty \le (D+1)/2$, at most $t(D+1)/2$. The switch multiplies the evaluation part of the noise by q'/q_B, and $\|\cdot\|_2 \le \sqrt{N}\|\cdot\|_\infty$. □

Corollary 7.8 (A proved B_ψ in the model)**.** *At the evaluation's PSI parameters (*$N_B = 2^{14}$*,* $t =$ 4,293,918,721*,* $D = 64$*,* $W = 7$ *windows and product trees of depth* $L_\times = 3$*, as in the implementation's balanced split of* d *by popcount,* q_B *of seven* 60*-bit limbs and* q' *of two), Proposition 7.7 gives* $E_0 = 2^{19.8}$*,* $E_3 = 2^{211}$*,* $E_{\mathrm{eval}} = 2^{262} < q_B/(2t) \approx 2^{387}$ *and* $\|w_{\mathrm{eval}}\|_2 \le 2^{-31}$ *for any* $E_{\mathrm{rl}}, E_{\mathrm{rns}} \le 2^{26}$*, and* 2^{-30} *for any* $E_{\mathrm{rl}}, E_{\mathrm{rns}} \le 2^{90}$ *(script* `experiments/proofs_check/bpsi_model.py`*). So, in the model, the evaluation part is covered by the "+1" of the implementation's* $B_{\mathrm{err}} = 2(\sqrt{N_B}/2 + N_B^{3/2}/2) + 1$*, and* $B_\psi := \sqrt{2\pi}B_{\mathrm{err}}$ *bounds* $\|w\|_2$ *with the rounding part (exact) and the fresh-encryption part (whose high-probability bound* $(q'/q_B)\beta_{\mathrm{fresh}}$ *is below* 2^{-250}*).*

Remark 7.9 (Status of B_ψ)**.** Proposition 7.7 is a theorem about the textbook BFV model (B1)–(B4); it is worst case over O's inputs and needs only the honest sampling of R's errors. That

OpenFHE 1.5.1's BFVrns (its RNS multiplication and scaling, and its relinearization) satisfies (B3) with small E_{rns} and E_{rl} was not checked; the measured evaluation noise after the switch is 2^{-72}, far below the model's 2^{-31}. G4's second layer is therefore proved for the model and remains conditional on this conformance for the implementation.

8 Correctness: no wraparound, and the exact sampler

Proposition 8.1 (Decryption without wraparound)**.** *Let $S_{\max} = \max_e \sup_D |\mathbf{z}_e(D)|$. Each entry is a sum over at most $\bar{n}_O$ rows of a product bounded by one of o, $\sqrt{ro}$, $\sqrt{rw}$, $\sqrt{ow}$ or w; so $S_{\max} \le \bar{n}_O \max(o, \sqrt{ro}, \sqrt{rw}, \sqrt{ow}, w)$. Let $d_{\max}$ bound $\|\widehat{Z}_s(D) - \Omega_{\text{fin}}\tau^{-1}(\mathbf{z}_s(D))\|_2$ over all D; Proposition 4.5 gives an explicit, proved $d_{\max}$ ($2^{30.3}$ at the evaluation setting with $\bar{n}_O = 10^6$ and $n_R^{\max} = 2{,}000$, and $2^{38.6}$ if every row may match). Suppose*

$$q_0 > 2\Big(\Omega_{\text{fin}}\,\tfrac{2|\mathcal{E}_s|}{N}\,S_{\max} + d_{\max} + \tfrac{1}{2} + B_{\text{tot},s} + t_\eta\,\sigma_{c,s}\Big). \tag{Q}$$

Then v_s, reduced to its centred representative modulo q_0, equals the integer polynomial $f_s(D) + e_{\text{tot},s} + \eta_s$. This holds except with probability at most

$$\delta_B + 2Ne^{-t_\eta^2/2}$$

over the keys and the fresh randomness. Moreover $\mathsf{Dcd}(v_s) = \tau(\widehat{Z}_s)/\Omega_{\text{fin}} + \tau(e_{\text{tot},s} + T_7' + \eta_s)/\Omega_{\text{fin}}$, where $T_7' = f_s - \widehat{Z}_s$.

Proof. Size of the centre. From the inversion formula in the proof of Lemma 1.1, $\|\tau^{-1}(\mathbf{z})\|_\infty \le \frac{2}{N}\|\mathbf{z}\|_1 \le \frac{2|\mathcal{E}|}{N} S_{\max}$. So $\|f_s\|_\infty \le \Omega_{\text{fin}}\frac{2|\mathcal{E}|}{N}S_{\max} + d_{\max} + \frac{1}{2}$.

Size of the error. $\|e_{\text{tot},s}\|_\infty \le \|e_{\text{tot},s}\|_2 \le B_{\text{tot},s}$, except with probability δ_B (Theorem 3.12).

Size of the noise. Each η_k is σ^2-subgaussian (Lemma 3.1(b)), so $\Pr[|\eta_k| > t_\eta\sigma] \le 2e^{-t_\eta^2/2}$. A union bound over the N coefficients gives the second term.

Under (Q), the integer polynomial $f_s + e_{\text{tot},s} + \eta_s$ lies strictly inside $(-q_0/2, q_0/2)^N$. So the centred reduction of v_s returns it exactly, and the decoding identity follows.

Earlier levels: no condition is needed. Proposition 3.8 is an identity of congruences that holds whatever the size of the intermediate plaintexts. A rescale satisfies $c_0' + c_1' s = (c_0 + c_1 s)/q + \varepsilon_0 + \varepsilon_1 s$ exactly in $\mathbb{Q}[X]/(X^N + 1)$; if $c_0 + c_1 s = \mu + KQ$ with K integral, whatever the size of μ, then $(c_0 + c_1 s)/q = \mu/q + K(Q/q)$, so the invariant modulo Q/q is $\mu/q + \varepsilon_0 + \varepsilon_1 s$. Plaintext products, additions and automorphisms are congruences modulo the current modulus, and so is key switching (Lemmas 3.5 and 4.1), again with no size condition; the rounding errors are bounded through their sup-norms only. Hence only the final reduction modulo q_0 can wrap around, and (Q) suffices. (This replaces the argued step of an earlier version, which bounded the intermediate plaintexts; the argument is that of the companion book, Chapter 9, Proposition `agg.wrap`, checked here.) □

Remark 8.2 (The specification's condition)**.** CALIBRATE requires

$$q_0 > 2\big(\Omega S_{\max} + 2^\kappa B_{\text{tot}} + 10\,\sigma_{c,\max}\big).$$

This implies (Q) with $t_\eta = 10$ when $\Omega = \Omega_{\text{fin}}$. The reasons:

- $2|\mathcal{E}|/N \le 1$, and the slack $\Omega_{\text{fin}} S_{\max}(1 - 2|\mathcal{E}|/N)$ is far larger than $d_{\max} + \frac{1}{2}$;
- $2^\kappa B_{\text{tot}} \ge B_{\text{tot}}$.

The specification should still state the correctness-failure probability $2Ne^{-50} \approx 2^{-57.1}$ per release. This is a correctness event, not a privacy event, because the view is the same whether or not wraparound occurs. The term $\Omega S_{\max}$ overestimates by a factor $N/(2|\mathcal{E}|) \approx 2^7$, so q_0 could be about 7 bits smaller.

The exact sampler. SampleDG is Algorithm 3 of Canonne, Kamath and Steinke [4]. It draws a discrete Laplace candidate Y with integer scale $t = \lfloor \sigma \rfloor + 1$, accepts it with probability $\exp(-(|Y| - \sigma^2/t)^2/(2\sigma^2))$, and repeats otherwise. The acceptance coins come from an exact Bernoulli($e^{-\gamma}$) sampler for rational γ. CKS20 prove the following, for rational σ^2:

- the output is *exactly* $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$;
- only integer arithmetic and uniform random bits are used;
- the expected number of iterations is bounded by a constant.

[*citation detail unverified:* the exact constant bounding the expected iterations in CKS20]

Two implementation requirements follow:

(i) $\sigma^2_{c,s}$ *must be rational.* Set $\sigma^2_{c,s} := \lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s) \rceil$. Increasing σ lowers ρ_s and only helps flooding.

(ii) *Big integers must be arbitrary-precision.* At $\sigma \approx 2^{55}$, t and σ^2 exceed 64 bits, so all arithmetic must be exact arbitrary-precision integer or rational arithmetic, and the coins must come from a CSPRNG.

What "constant-time" must mean. The model of the specification abstracts side channels away. Its remaining obligation O2 nevertheless asks for a constant-time sampler, and R can at least observe O's response time. The requirement is therefore:

> (CT) The joint distribution of everything R can observe about the execution of the release, in particular its duration, is independent of (η, s_O, u, e_0, e_1) given the public parameters.

The rejection structure helps only in part:

- In rejection sampling, the number of *rejected* trials is independent of the accepted value.
- But the time of the accepting trial depends on the accepted Y. The discrete Laplace and Bernoulli($e^{-\gamma}$) subroutines loop for value-dependent numbers of iterations, and big-integer operations take time that depends on operand length.

So a direct implementation of CKS20 does not satisfy (CT).

A sufficient implementation has four parts:

(a) every inner loop runs for a fixed public number of iterations $T_{\max}$, with the result selected branch-free;

(b) the probability that some loop would have needed more than $T_{\max}$ iterations is at most $2^{-\kappa'}$ per sample, and the total $Nn_{\mathrm{rel}}2^{-\kappa'}$ is added to the slack γ;

(c) all arithmetic is done in fixed-width limbs;

(d) the N samples of a release are produced in one batch whose duration is padded to a public bound.

A concrete variant (Section 5.5). The implementation now uses SampleDGBudget (Definition 5.14), which realizes (a)–(c) in a stronger form: the whole batch, not each loop, gets a fixed budget of T candidate trials; every trial executes the same fixed-width operations (limb arithmetic, every coin of the bounded loops drawn and compared), the surplus is discarded, and the call fails with probability at most ξ_T (Lemma 5.17) instead of $Nn_{\mathrm{rel}}2^{-\kappa'}$; $n_{\mathrm{rel}}\xi_T$ is added to γ (Corollary 5.21). It is exact on the complement of the failure event (Lemma 5.16), so no Rényi or statistical distance of the output enters, unlike generic constant-time samplers by convolution such as that of Micciancio and Walter [14]. (d) is kept as defence in depth. (CT) then holds for η up to ξ_T per release under the cost assumption (W) (Assumption 5.18). **[Not fully rigorous:** (W) is not verified for the implementation's CPython and numpy operations (Remark 5.24), and (CT) for (u, e_0, e_1) depends on OpenFHE's samplers, which were not analysed.]

9 Calibrate, the run as a secure computation, and O's work

The three results of this section were first written in the companion book (Chapter 19, Proposition `par.calibrate`; Chapter 4, Proposition `sc.secureDP` with Chapter 17, Theorem `run.secure`; Chapter 20, Proposition `cost.lower`). They were checked here, with one correction to the second.

9.1 Soundness of Calibrate

CALIBRATE (the paper's Section on calibration) takes the parameter vector $\mathcal{P}$ and the public ledger parameters and runs seven steps: the security test $\log_2(QP) \le L_{\max}(N, \lambda_{\text{sec}})$; the sensitivities and the mode; $\kappa = \max\{\kappa_{\min}, \kappa_{\text{run}}, \kappa_L\}$ (unless $\mathcal{P}$ fixes κ), with the test of the run's share of the ledger's slack capacity; γ and δ_{DP}; ρ_{run} and its split, the stage-2 share rounded down; the smallest grid scale at which $\sigma_{c,s} = (\lceil \widehat{\Delta}^2_{\text{enc},s}/(2\rho_s) \rceil)^{1/2} \ge \max\{2^\kappa B_{\text{tot},s}/\sqrt{2\pi}, 2^{42}\}$ for every release; and the test on q_0. It refuses when a step has no solution.

Proposition 9.1 (CALIBRATE is sound)**.** *Suppose* CALIBRATE *does not refuse, and that it evaluates $B_{\text{tot},s}$ and $\Delta_{x,s}$ by the formulas of Theorem 3.12 with Lemma 4.1, and of Lemma 4.3, for the circuit model in use. Then:*

(a) every output is a deterministic function of $\mathcal{P}$ and the public ledger parameters;

(b) for every release, $\sigma_{c,s} \ge \max\{2^\kappa B_{\text{tot},s}/\sqrt{2\pi}, 2^{42}\}$, $\widehat{\Delta}^2_{\text{enc},s}/(2\sigma^2_{c,s}) \le \rho_s$ with $\sigma^2_{c,s}$ an integer, $\sum_s \rho_s \le \rho_{\text{run}}$, $\rho_{\text{run}} = \max\{\rho : \delta(\rho, \varepsilon_{\text{run}}) \le \delta_{\text{DP}}\}$ with $\delta_{\text{DP}} = \delta_{\text{run}} - (1 + e^{\varepsilon_{\text{run}}})\gamma > 0$; these are the hypotheses of Theorem 5.11;

(c) condition (Q) of Proposition 8.1 holds for every release, with $d_{\max}$ of Proposition 4.5, and the chain meets the security bound;

(d) if κ is derived, $(1 + e^{\varepsilon_{\text{run}}})\gamma \le \zeta\delta_{\text{run}}$ and $\rho_{\text{run}} \ge \tilde{\rho}_{\text{run}}$; with a ledger cap, $n_{\text{rel}}\varepsilon'(\kappa) \le (\rho_{\text{run}}/\rho_L)E'_L$ and $n_{\text{rel}}(\delta_B + \delta') \le (\rho_{\text{run}}/\rho_L)\Gamma_L$, and the costs it returns are the per-run costs of the composition appendix (Part C, Section 5).

Proof. (a) No step reads data, keys, errors or coins: $B_{\text{tot},s}$ and $\Delta_{x,s}$ are functions of public parameters only (Theorem 3.12, Lemmas 4.1 and 4.3), the grid and its order are fixed, and every step is deterministic. (b) The scale loop stops only where the flooding condition holds for every release. $\sigma^2_{c,s} = \lceil \widehat{\Delta}^2_{\text{enc},s}/(2\rho_s) \rceil \ge \widehat{\Delta}^2_{\text{enc},s}/(2\rho_s)$. The split sums to at most ρ_{run} because the stage-2 share is rounded down. $\delta_{\text{DP}} > 0$ by the test, and ρ_{run} exists and is positive because $\delta(\cdot, \varepsilon)$ is continuous, increasing and 0 at 0. (c) The q_0 test is $q_0 > 2(\Omega_{\text{fin}} S_{\max} + 2^\kappa B_{\text{tot},\max} + t_\eta \sigma_{c,\max})$ with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$. Since $\Omega_{\text{fin}} S_{\max} = \Omega_{\text{fin}} \frac{2|\mathcal{E}_s|}{N} S_{\max} + (1 - \frac{2|\mathcal{E}_s|}{N})\Omega_{\text{fin}} S_{\max}$ and the second part exceeds $d_{\max} + \frac{1}{2}$ whenever $d_{\max} + \frac{1}{2} \le (1 - 2|\mathcal{E}_s|/N)\Omega_{\text{fin}} S_{\max}$ (at the evaluation setting $2^{38.6}$ against more than 2^{74}), and $2^\kappa B_{\text{tot},\max} \ge B_{\text{tot},s}$, $\sigma_{c,\max} \ge \sigma_{c,s}$, the test implies (Q). (d) $\gamma(k)$ is strictly decreasing, so $\kappa \ge \kappa_{\text{run}}$ gives the share and $\delta_{\text{DP}} \ge (1 - \zeta)\delta_{\text{run}}$, hence $\rho_{\text{run}} \ge \tilde{\rho}_{\text{run}}$ by monotonicity; $\varepsilon'(k)$ is decreasing in k, so $\kappa \ge \kappa_L$ and $\tilde{\rho}_{\text{run}} \le \rho_{\text{run}}$ give the smudging inequality, and the share test gives the slack inequality. □

Remark 9.2 (Second reading)**.** The proof is in exact arithmetic. The implementation computes $\sigma^2_{c,s}$ as an exact fraction, rounds the split and the ledger's capacities in the safe direction, and stores the ledger's sums exactly; it evaluates $B_{\text{tot},s}$ in double precision (relative error about 10^{-15}) and, since 2026-10-25, makes the flooding test conservative: $B_{\text{tot},s}$ and the needed $\sigma_{c,s}$ are rounded up by a relative 2^{-40}, and the integer $\sigma^2_{c,s}$ is compared with the square of the rounded value in exact rational arithmetic, so floating-point error cannot pass a case that exact arithmetic would refuse. Its formula for the term T_8 uses d_{num} where $\sqrt{d_{\text{num}}}$ suffices, which is conservative. Its diagnostic value `q0_need_Q` used $d_{\max} = 2^{35}$ until 2026-10-25, which Proposition 4.5 proves when at most

$n_R^{\max} = 2{,}000$ rows match ($2^{30.3}$) but not when every row of a register of 10^6 may match ($2^{38.6}$). It now uses the $d_{\max}$ of Proposition 4.5 for $n_{\max} = \min(n_R^{\max}, \bar{n}_O)$; the condition (Q) still needs 74.1 bits at $\bar{n}_O = 4{,}096$ and 81.9 bits at 10^6 ($\varepsilon = 1$), because its dominant term is about 2^{81}. The refusal test itself uses the condition of (c), which needs no $d_{\max}$.

9.2 The run computes the DP functionality against R

Let $\mathcal{M}(D) = (\omega_{\mathsf{R}}, \Lambda, ([f_s(D;\pi) + \eta_s]_{q_0})_s)$ be the mechanism of Theorem 5.11, where $[\cdot]_{q_0}$ is the centred residue modulo q_0 (a post-processing, so $\mathcal{M}$ is still ρ_{run}-zCDP), and let Out be R's output map: its decryption of the Link replies and the centred residues $v_s = h_{O,s} + c'_{1,s} s_R \bmod q_0$.

Corollary 9.3 (Standard simulation for R, jointly with its output)**.** *Under the hypotheses of Theorem 5.11, the pair (R's view, R's output) is within statistical distance γ plus computational loss $n_{\text{rel}}\text{Adv}_1 + \text{Adv}_2$ of $(\mathcal{S}(\mathcal{M}(D)), \mathcal{M}(D))$: the run computes the functionality that hands $\mathcal{M}(D)$ to R, against a semi-honest R, in the standard (not the liberal) sense. With Theorem 7.3 in place of the transcript Λ, the functionality may output $(I, \pi|_I)$ instead of Λ, at the additional cost of that theorem.*

Proof. Out is an efficient deterministic function of the view, and the simulator of Theorem 5.11 is output-consistent: on input w_s it outputs $(c^*_{1,s}, w_s - c^*_{1,s} s_R \bmod q_0)$, whose reconstruction $h^* + c^* s_R \bmod q_0$, taken as a centred residue, is w_s exactly because $w_s = [f_s + \eta_s]_{q_0}$ is already reduced. Applying $v \mapsto (v, \text{Out}(v))$ to the two views of Theorem 5.11 preserves both the statistical and the computational distance, and maps the simulated pair to $(\mathcal{S}(\mathcal{M}(D)), \mathcal{M}(D))$. □

Remark 9.4 (Correction to the book)**.** The book's mechanism outputs $f_s + \eta_s$ unreduced, and its claim that the reconstruction returns it "exactly" fails on the wraparound event $\|f_s + \eta_s\|_\infty \geq q_0/2$, of probability at most $2Ne^{-t_\eta^2/2}$ per release under (Q). Either the functionality outputs the centred residue, as above (nothing changes for privacy), or that probability is added to the slack. The book's general statement (SIM-CDP with an output-consistent simulator implies security jointly with the outputs, and the converse fails without output-consistency) is correct as written.

9.3 O's work is linear in its scope

Model O's computation as reading cells of its register (one cell per row), its final state as a function of its coins and the cells read, and the decrypted statistic as a function of the final states; the protocol is $(\tau_{\text{dec}}, p_{\text{err}})$-correct if, for every input in the clipped domain, the decrypted $\mathbf{z}$ is within τ_{dec} of the true one in every entry except with probability p_{err}.

Proposition 9.5 (Work linear in the scope)**.** *Let the protocol be $(\tau_{\text{dec}}, p_{\text{err}})$-correct with $B_O^2 > 2\tau_{\text{dec}}$, and let every efficient distinguisher of O's views for two cohorts of the same size have advantage at most* adv*. If n_{Sc} rows of the scope register are non-dummy, each matchable by some cohort, then O reads at least $(1 - 2p_{\text{err}} - \text{adv})\, n_{\text{Sc}}$ rows in expectation, for every input.*

Proof. Fix a non-dummy row t and let E_t be the event that cell t is never read; it is a function of O's view. If t is matched, couple two executions with the same coins and inputs except $\mathbf{x}_{O,t} = \mathbf{0}$ in one and $B_O \mathbf{e}_1$ in the other. They coincide until cell t is read, so E_t has the same probability in both, and on E_t the decrypted statistics coincide while the true entries $\mathbf{B}_{11}$ differ by $B_O^2 > 2\tau_{\text{dec}}$; so on E_t at least one execution errs, and $\Pr[E_t] \leq 2p_{\text{err}}$, for every content of the other cells. If t is unmatched, replace one identifier of the cohort by that of row t; the indicator of E_t is an efficient function of O's view, so its probability changes by at most adv. Summing $\Pr[t \text{ read}] \geq 1 - 2p_{\text{err}} - \text{adv}$ over the n_{Sc} rows gives the bound. □

In v3.1, adv is the slack of G2 (Corollary 6.3), so O's work is $\Omega(n_{\mathrm{Sc}})$; O may skip its dummy rows, which it knows, so the bound is in n_{Sc}, not $\bar{n}_O$. The argument is that of the classical lower bound on the server's work in single-server private information retrieval.

10 Corrections to the specification

Items are ordered by importance. "Spec" refers to `protocol_v31.tex`.

Correction 1 (The centre of the mechanism; affects Lemma 1, Theorem 1 and Calibrate)**.** *Spec:* the ideal mechanism is $\bar{\mathbf{z}}_s(D) + \eta_s$ with $\bar{\mathbf{z}}_s = \mathsf{Ecd}(\mathbf{z}_s(D))$, and $e_{M,s}$ includes O's encoding roundings of V_e and $\mathbf{u}_k$.

Problem: the decrypted value differs from $\mathsf{Ecd}(\mathbf{z}_s(D))$ by deterministic, data-dependent terms. Chief among them is the mask spill $\sum_e S_e\, r_{u_e}$, of norm about $2^{30.7}$ to $2^{34.5}$ for $\bar{n}_O = 10^6$ (Remark 3.9). Flooding them would require $\log_2 \Omega \approx 67$ to 71.

Corrected statement: centre the mechanism at the noiseless circuit output $f_s(D) = \lfloor \widehat{Z}_s(D) \rceil$ (Definition 3.7), and use the sensitivity

$$\widehat{\Delta}_{\mathrm{enc},s} = \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N}$$

(Corollary 3.13), where $\Delta_{x,2} \approx 2^{18.8}$. Then $e_{\mathrm{tot},s}$ consists only of randomness-dependent terms (Proposition 3.8), and Theorem 5.11 holds with $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$. Numerically this changes ρ_s by a relative $2^{-32.7}$. Lemma 1.2 itself is correct as stated.

Correction 2 (Exact final scale)**.** Throughout Lemma 1, Calibrate, σ_s and Dcd, Ω must be the exact final scale $\Omega_{\mathrm{fin}} = \Omega_U\Omega_V\Omega_u/(q_1q_2)$. It must be tracked exactly, as in FLEXIBLEAUTO, or computed from the primes. Decoding with a nominal power-of-two scale, as FIXEDMANUAL does, causes a deterministic relative bias of about 2^{-30}; we measured $8.7 \cdot 10^{-10}$ (Section 3.6). This bias is not a privacy problem once the mechanism is centred at f_s, but it is an avoidable estimation bias.

Correction 3 (Obligation O1: the value of B_{tot} and the required scale)**.** *Spec:* $\log_2 \Omega \approx 52.5$ at $\varepsilon = 1$.

Problem: that figure floods only the fresh-encryption error, and only at its mean ℓ_2 norm. For the circuit of Algorithm 4 as written, Theorem 3.12 gives a rigorous worst-case bound of $B_{\mathrm{tot}} \approx 2^{25.3}$ for stage 2, with $\delta_B = 2^{-50}$. That requires $\log_2 \Omega \approx 61.5$, or 60.5 if $P \geq 2^{20}\tilde{Q}_{\max}$ (Table 1).

Recommended changes: the following give $B_{\mathrm{tot}} \approx 2^{17.6}$ and $\log_2 \Omega \approx 53.7$:

(i) evaluate EvalSum before the first rescale, at level 2;
(ii) sum the masked products of a release *before* a single second rescale, instead of rescaling each P_e;
(iii) use stochastic rounding in Rescale and in ModDown;
(iv) choose $P \geq 2^{10}\tilde{Q}_{\max}$;
(v) encode R's vectors at $\Omega_U = q_2 \approx 2^{60}$.

By a rough count, $\log_2(QP) \approx 320$ for (A) and ≈ 285 for (C′). Both are below the 128-bit limit of about 438 for $N = 2^{14}$ [2]. [**Not fully rigorous:** This count is approximate; the modulus chain should be re-derived with the chosen d_{num}.]

Further requirement: Calibrate must use the *worst-case*, data-independent $B_{\mathrm{tot},s}(\Omega)$ of Theorem 3.12. A value estimated from the actual ciphertexts, for example by OpenFHE's noise estimation, would depend on the data. It would then leak through $\sigma_{c,s}$, and it would not be a worst-case bound. Such an estimate may serve as a sanity check only.

Correction 4 (Lemma 2: constants)**.** *Corrected statement:* for $e \in \mathbb{Z}^N$ and $\eta \sim D_{\mathbb{Z}^N,\sigma_c}$ with $\sigma_c > \sigma_2 \geq \eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi}$,

$$\mathrm{SD}(\eta + e, \eta) \leq \frac{\|e\|_2}{\sqrt{2\pi}\sqrt{\sigma_c^2 - \sigma_2^2}} + \frac{4\epsilon}{1-\epsilon}.$$

Alternatively, $\mathrm{SD}(\eta + e, \eta) \leq \|e\|_2/(2\sigma_c)$ for every $\sigma_c > 0$. With CALIBRATE's condition, the per-release distance is at most $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ (Lemma 2.3, Corollary 2.4).

Correction 5 (δ accounting)**.** *Spec:* $\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - n_{\mathrm{rel}}(\delta_B + 2^{-\kappa})$.

Corrected statement: Theorem 5.11 gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP with statistical slack

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big),$$

plus the sampler-cap failures of Section 8. A single indistinguishability guarantee $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$ requires

$$\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$$

(Lemma 5.7). The factor $1 + e^{\varepsilon}$ is missing in the spec. For the example ($\gamma \approx 2.7 \cdot 10^{-12}$) the effect on ρ_{run} is negligible.

Correction 6 (Assumptions of Theorem 1)**.**

- *RLWE in two forms.* RLWE is needed both for the re-randomizer (one sample per release, at modulus q_0, with ternary u and $\mathcal{N}_{\mathbb{Z}}(0, 3.2^2)$ errors) and for O's key shares (multi-sample, at modulus PQ_2, with circular security for the rotation-key shares). These are Assumptions 5.9 and 5.10.
- *What "threshold key-generation security" means here.* For the one-round protocols of [17] it is exactly Assumption 5.10.
- *Unneeded assumptions.* PSI security is *not* needed for G3 under the spec's adjacency, because the identifiers are fixed and the PSI transcript is independent of $(\mathbf{x}_O, y)$. Nor is the random-oracle model.
- *An implicit requirement.* CALIBRATE's outputs must be data-independent.

Correction 7 (PSI and the disclosure claim G4)**.** G4 says that R learns only I and $\pi|_I$ from the Link phase. This requires the labeled PSI to be secure against a semi-honest receiver. For the FHE-based PSI of [6, 5], that requires *circuit privacy* of O's BFV replies, for example by noise flooding or re-randomization before the reply. Without it, the reply noise depends on O's identifier set. A modulus switch alone (Algorithm 2, line 4) is a heuristic measure. *Update:* with OPRF-keyed items and labels, the first layer of G4 holds without circuit privacy, in the random-oracle model under CDH (Section 7, Theorem 7.3); SANITIZEBFV adds a statistical second layer, proved for a model of BFV (Proposition 7.7). [*citation detail unverified:* how CLR17 and CHLR18 phrase the noise-flooding requirement]

Correction 8 (G1 and G2: assumptions and slack)**.** *Missing assumptions:* G1 needs Assumption 6.1:

- RLWE with circular security in s_R, covering R's rotation-key shares;
- the random oracle for the seed expansion;
- IND-CPA of the PSI's BFV, with circular security for any relinearization key;
- obliviousness of the OPRF.

Simulator inputs: the simulator also takes the public parameters and O's own input.

Slack: G1 and G2 hold up to the probability p_{cuckoo} that R's cuckoo hashing fails observably.

Correction 9 (Correctness)**.** The q_0 condition in CALIBRATE is sufficient once $\Omega = \Omega_{\mathrm{fin}}$ (Proposition 8.1). The spec should state the per-release correctness-failure probability, $\delta_B + 2Ne^{-50} \approx 2^{-50}$.

The sharper condition (Q) saves about 7 bits of q_0. The "modulus switch to q_0" in Algorithm 4 is a no-op, or an exact limb drop, for a chain $q_0 q_1 q_2$. A scaling modulus switch would add a rounding term bounded like T_5.

Correction 10 (Sampler)**.**

- Set $\sigma_{c,s}^2$ to a rational, for example $\lceil \widehat{\Delta}_{\mathrm{enc},s}^2/(2\rho_s) \rceil$, as CKS20's exact sampler requires.
- "Constant-time" must mean (CT) of Section 8. A direct implementation of CKS20 does not satisfy (CT). The budgeted sampler of Section 5.5 does for η, under the cost assumption (W), up to a failure probability ξ_T per release that is added to γ.
- The slot-domain noise has variance at most $\sigma_{c,s}^2 N/(2\Omega_{\mathrm{fin}}^2)$ per real slot component. The variance of $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ is below σ^2, and at $\sigma \gg 1$ the difference is negligible. So $\sigma_s = \sigma_{c,s}\sqrt{N/2}/\Omega_{\mathrm{fin}}$ is an upper bound on the noise s.d., not an exact value.

Remark 10.1 (Statements verified as correct)**.** The following claims of the spec were checked and need no correction:

- the identity $\|\tau^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\|\mathbf{z}\|_2$ and the rounding term $\sqrt{N}$ of Lemma 1;
- the zCDP cost $\rho_s = \Delta_{\mathrm{enc},s}^2/(2\sigma_{c,s}^2)$ of the lattice discrete Gaussian, once Δ_{enc} is replaced by $\widehat{\Delta}_{\mathrm{enc}}$;
- the validity of the conversion used in CALIBRATE;
- the release format (c_1', h_O) with c_0' withheld, which removes the C3 leak;
- the need for fresh (u, e_0, e_1, η) in every release;
- the absence of any decryption share from R.

The spec's remark that stage 1's error "cannot depend on the entries released only in stage 2" is true but not needed: flooding covers any correlation.

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